Absolute Calibration of ALIS Cameras

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Abstract

This master thesis focuses on Auroral Large Imaging System (ALIS) image pre-processing and finding the physical value for ALIS imagers output. The primary scientific objective of ALIS is to investigate auroral physics, also the airglow and other atmospheric phenomenon. Thus, to know the physical meaning of imagers output is the first and most important task in a complete analysis of ALIS data.

This report presents detailed procedures for ALIS image pre-processing and absolute calibration by using IRF laboratory data from Sept. 2007. Non linearity issues of the pixel response are also discussed and correction functions are suggested. Results from 5 ALIS imagers are listed in the report.

To investigate how atmosphere and camera domes affect camera sensitivity values, it presents a comparison between absolute calibration results by using laboratory data and Pulkovo spectrophotometric catalog, and different has been found and discussed.

Keywords: ALIS, Image pre-processing, Absolute calibration.

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Chapter 1

Introduction

> *"We will make it run the dynamos to supply our houses and streets with electric light;… and it shall develop the brains of our statesmen and legislators, to make them wiser and better and of more practical use than they are at present. Wonderful things will be done when we get the electricity of the aurora under our control."*
> *~Thomas W. Knox, The Voyage of the Vivan, 188んあわ4*

An aurora is a colorful and dynamic natural phenomenon displayed in the night sky in the Polar Regions, which is caused by a complicated interaction between the solar wind and terrestrial atmosphere. Different from other natural phenomena in the troposphere, such as tornados, thunder storms, or rainbows, an aurora results from photon emission in the thermosphere – the upper atmosphere whose lower boundary is usually 80km to 120km above the earth's surface. Because the Earth is continually affected by solar wind -- a stream of hot plasma emitted by the sun consists of electron and proton usually with energies from 1.5 to 10 KeV, the magnetosphere is full of the charged particles, which are trapped by strong magnetic field as the solar wind passes the earth. These particles confined by electromagnetic force, moving along magnetic field lines and eventually accelerated toward the earth, are responsible for exciting atoms in the upper atmosphere, and excitation energy is lost by emission of photons when atoms returning from excited state to ground state.

Auroral studying helps us know more about Earth's surrounding space weather and the atmosphere. During the past several decades, auroral studying has developed from an immature theory into a more precise scientific discipline central in the study of the upper atmosphere, particularly in ionosphere and magnetosphere. However, many questions are still remained, for example, in auroral morphology, the reason why the aurora to appear as thin, the shape of curtain is still unclear. Thus, along with developing of observation techniques, more and more researches should be focus on this magnetospheric process.

In common, scientists always study aurora in three dimensions: rockets and satellite borne observations, equipments on jet aircraft and ground based observatories. Due to the merits of continuous studies and less spatial smearing, which can not be achieved by air or space borne instruments because of orbital motion, a large part of our knowledge of the behavior of aurora has been obtained by observation using ground based detectors sensitive to a wide range of wavelengths. Since aurora is primarily an optical phenomenon, auroral imaging is one of the most important techniques to observe it. Several designs of all-sky camera systems appeared in the 1950s, before *International Geophysical Years* (1958-1959). The merits of multi-station imaging are (1) its capability to determine the position and motion of imaged objects and (2) the possibility to estimate the inner distribution of apparent objects by using tomographic methods (*Gustavsson*, 2000). In history, optical multi-station measurements were investigated by *Romick and Belon* (1967), and with a further step of general tomographic inversion made by *Solomon* (1984). Because much more sensitive detectors exist today, a re-examination of spectral features of the aurora is desired.

ALIS suggested by *Steen* (1989) and described by *Brändström* (*2003*) is a ground-based optical system for measuring aurora, high-altitude clouds, and other atmospheric optical objects. ALIS is highly configurable and many configuration options exist to enable different operational modes. Each imagers has a filter wheel with space for up to six narrow band interference filters One of the main tasks of ALIS is to determine the 3-D spatial structures of aurora. The procedures to determine what physical value each pixel represents, so called camera absolute calibration, is the first and the most important task before starting analysis of ALIS data. This thesis presents detailed procedures about this round of absolute calibration by using laboratory data from Sept. 2007, furthermore, in order to know how the natural environment affects our data from aurora and airglow, stellar calibration is carried out in this thesis.

This thesis is organized into 6 chapters. Chapter 2 gives a short description about the ALIS stations and imagers, to provide readers with a basic understanding of ALIS. Chapter 3 gives a short description about the geometrical calibration of ALIS, which is

developed by *Gustavsson*, (2000). Camera optical parameters are calculated by matching night sky images to a brightness star catalogue, and a pixel field-of-view is derived in a general, straightforward way. Chapter 4 presents work about removing CCD signatures and proposes a camera non-linear correction method. ALIS CCD absolute calibration with the laboratory data is carried out. Chapter 5 presents the stellar absolute calibration of camera 3 and camera 5. Finally, chapter 6 concludes this thesis and proposes possible future work.

Chapter 2

ALIS stations and imagers

2.1. ALIS stations

The ALIS (Auroral Large Imaging System) project was first proposed by *Steen* (1989), and a detailed description is covered in (*Steen and Brändström*, 1993 and *Brändström, 2003*). It is a ground-based optical system for measuring the aurora, high-altitude clouds, airglow, and other atmospheric phenomenon. In order to fulfill the requirement for pixel line of sight accuracy, it consists of a grid of observation with approximately 50km separation. The first proposed layout of ALIS provided for 28 stations (*Steen*, 1989), and until today six of them have been built and are in regular unmanned operation.

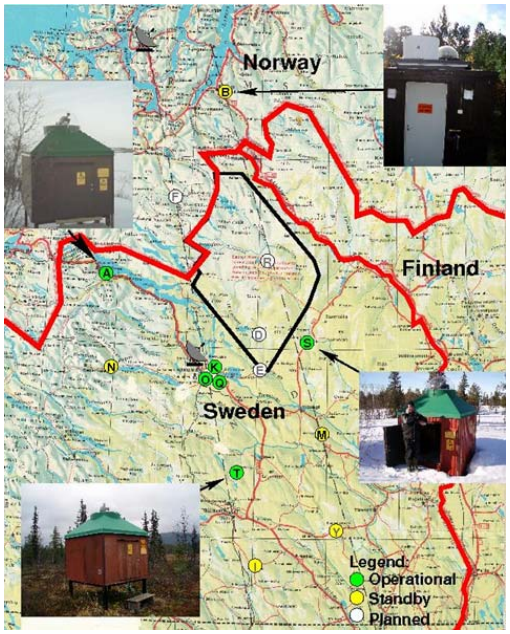


Figure 2-1 Map of ALIS stations

In order to observe four auroral lines and improve discrimination against moonlight and artificial lights, narrow band interference filters ($\Delta\lambda = 50$ Å for 4278 Å and $\Delta\lambda = 40$ Å for 5577 Å, 6300 Å and 8446 Å) have been mounted in a filter wheel with six positions. Ångström is a unit of length (*Ångström*, 1868). Four spectral lines from the aurora and airglow measured by ALIS are described by *Gustavsson* (2000):

4278Å, blue line is caused by transition of the molecular nitrogen ion from state $N_2^+\left(B^2\Sigma_u^+\right)$ to the first vibrational level $N_2^+\left(X^2\Sigma_u^+\right)$ of the electronic ground state.

5577Å, green line is emitted by atomic oxygen as it transitions from the second lowest excited state $O\left(^1S\right)$ to the lowest excited state $O\left(^1D\right)$.

6300Å, red line is emitted by atomic oxygen transferred from its lowest excited electronic level $O\left(^1D\right)$ to the atomic ground level $O\left(^3P_2\right)$.

8446Å, near infrared line results from the transition $O\left(3p^3P\right) \rightarrow O\left(3s^3S\right)$.

Only three lines (4278Å, 5577Å and 6300Å) are mentioned and have been calibrated in this thesis as no laboratory calibration exists for line 8446 Å so far.

2.2. ALIS imagers and data

2.2.1. ALIS imagers

ICCDs were considered early for the ALIS project (*Steen*, 1989). However, after a series of scientific considerations, a thinned back-side illuminated quad-read-out CCD was selected for ALIS because of better linearity and has long-time stability without the blooming effect which is caused by an image intensifier. And, for ALIS imagers, SNR can be enhanced by increasing pixel area also by increasing integration time (*Brändström*, 2003). Utilizing on-chip binning factors could increase pixel area and time resolution; meanwhile, spatial resolution is decreased. An optimization compromise can be made between sensitivity, spatial and time resolution by binning operation. To achieve the threshold of satisfactory SNR in low light conditions (column emission rate down to a couple of hundred Rayleigh), a common shortest integration time is about 1 second.

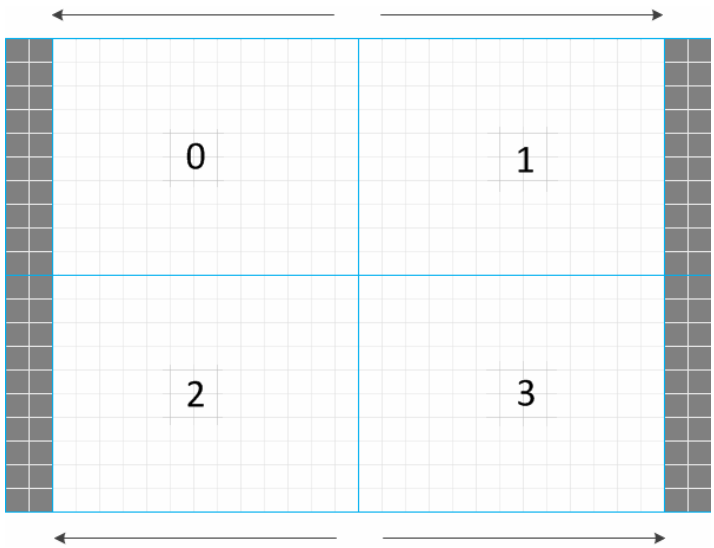


Figure 2-2 Layout of sub arrays of ALIS imagers, quadrants are denoted 0-3 and totally in 1024 by 1024 pixels. The shaded areas at the two sides of CCD chips indicate the bias pixels, also called over-scan strip or reference-pixels which are shaded from illumination. The area of bias pixels are slight different according to different Camera.

The formula to calculate frame rate is a function of integration time, read-out time and misc. time before next read (*Brändström*, 2003).

$$FR = \frac{1}{t_{\text{int}} + t_{read} + t_{misc}} \tag{2.1}$$

A suitable compromise is required between signal-to-noise ratio and frame-rate. It is already known that the read-out noise increases as the square root of the pixel read-out frequency. For ALIS imagers, measured pixel read-out rate is approximately 10.5 μs /pixel, thus for the case binning 1×1, it costs around 11s to read 1024×1024 pixels (*Brändström*, 2003). At a given pixel rate, one method to increase frame-rate is to divide the CCD into sub-arrays. Figure 2-2 shows the ALIS CCD is divided into four sub arrays with identical read-out channels working in parallel. Meanwhile, due to the bias frame of four quadrants is unequal, dividing CCD chips into sub-arrays causes "inhomogeneous" in raw ALIS data, i.e. the pixels' value varies dramatically at the quadrant edge (see Figure 2-3), and this "step" variation must be removed during the ALIS image pre-processing procedures.

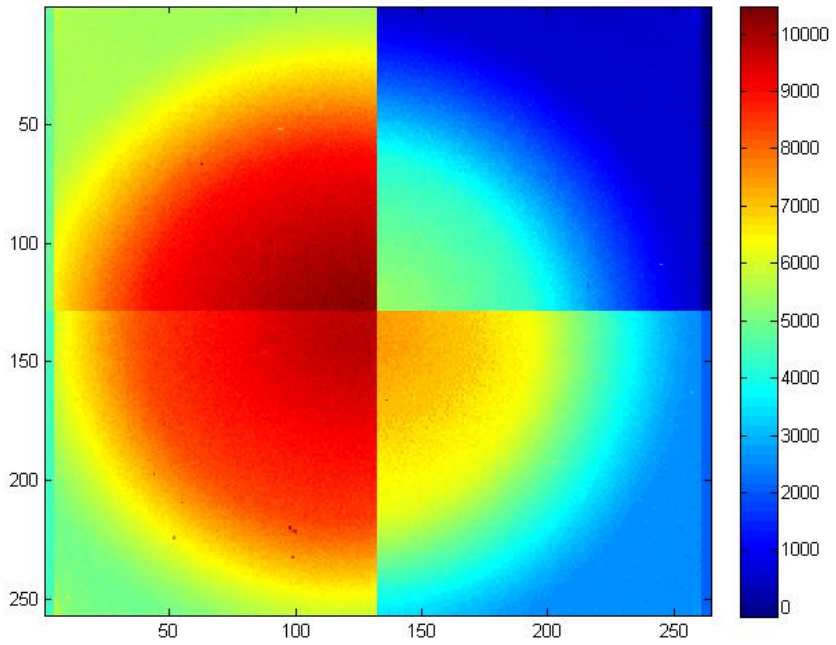


Figure 2-3 Raw laboratory data from Camera 5, Bin 4by4, Gain 13

2.2.2. ALIS data

All ALIS data are available on web site (http:// www.alis.irf.se), and are stored in FITS (Flexible Image Transfer System) format. Information regarding an image (exposure time, binning, gain, CCD number, filter, CCD operation temperature etc) can be found in the header of each file.

For each imager, images taken at different binning, gain, and exposure time are stored in a same folder, by taking advantage of the key words in the file header, arbitrary images with needed properties could be collected from thousands of ALIS data. For Example, due to we need more than 50 images taken by same binning, gain and zero exposure time to calculate bias frame for each imager (See chapter 4), it makes us more convenient to cluster such a set of images by searching the key words 'INSTRUME', 'ALISGTBL', 'ALISXBIN', 'EXPTIME', and then classify them with the same values.

2.3. Summary

Because an extensive, detailed description about ALIS is covered in (*Brändström*, 2003), this chapter just provides an introductory overview of ALIS, gives readers a basic knowledge about ALIS and makes it easier to understand the major task in the following chapters.

Chapter 3

Geometrical calibration of ALIS

3.1. Introduction

It is physically meaningful to know the number of photons which fall into the field-of-view of a pixel and which hit its effective area to produce a count. An ALIS CCD outputs two-dimensional raw image with size 1024×1024 pixels and a signed 16-bit integer value is used to present the raw counts produced by incoming photons for each pixel in exposure time.

With knowledge of the number of photons per count, radiance in quantum unit can be derived from raw counts, defined as photons per solid angle per square meter per second, by using formula (*Gustavsson*, 2000):

$$L_\gamma = \frac{\mathbb{N}}{d\Omega A_L d\lambda t}\left[\frac{photons}{m^2 \cdot sr \cdot s}\right] \tag{3.1}$$

Where,

$\mathbb{N}$: Number of photons detected by pixel with field-of-view,

$d\Omega$: Solid angle of pixel field-of-view,

A_L : Pixel effective area,

$d\lambda$: Wavelength band of photons,

t : Exposure time.

As the regards of the interpretation in terms of physical process should be convenient, in aeronomy, Hunten proposed that photometric measurements of aurora and airglow be reported in terms of $4\pi L_\gamma$, rather than radiance L_γ, further it is has been given the name Rayleigh (*Hunten*, 1956),

$$1[Rayleigh] \equiv 1[R] \triangleq 10^{10}\left[\frac{photons}{m^2 \cdot s \cdot column}\right] \tag{3.2}$$

The word column denotes the concept of an emission-rate from a column of unspecified length (*Baker and Romick*, 1976).

For ALIS absolute calibration, the task is to find the pixel sensitivity value, $\frac{[Rayleighs]}{[Count]}$.

One of the reasons cause slightly variations of pixel-to-pixel sensitivity is the nonuniform field-of-view and effective area of CCD pixels. Thus, correction should be carried out during the procedure of absolute calibration.

3.2. Geometrical calibration

Similar to the procedure of image registration, CCD geometrical calibration is carried out to determine the transformation between object and image space. First of all, to fulfill the accuracy requirement, selecting a good camera model is quite essential and important and then finding the unknown parameters of the camera model is our main task. Because of the high accuracy of angular positions, star catalogue is particularly adequate for the astronomical camera's geometrical calibration purpose. Method of exploiting fixed stars as interest points was originated in the 1974, which is used for calibrating the Orbigon lens (*Schmid,* 1974). Also some researches have done this by using the techniques of tracking stars, which provide high precision attitude information for spacecraft or distortion parameters for a universal camera (For example*, Bak, Wisniewski and Blanke*, 1996, *Klaus and Bauer*, 2004).

For ALIS, Gustavsson presented a well done method to determine the camera's orientation and optical parameters of imagers system (*Gustavsson,* 2000). With the knowledge of those optical parameters, field-of-view of pixels and effective area can be derived directly.

3.2.1. Optical parameters

Optical parameters are eight values as the following: optical model focal widths in horizontal and vertical direction, camera rotation angles (α, β, γ), image coordinates (relative units) for projection point of optical axis (x and y direction) and shape factor. With the knowledge of these eight parameters, we can calculate the optical axis point direction and object projection positions in image space. Furthermore, according to these values, it is possible to determine the field-of-view of each pixel.

To avoid the radial error for the pin-hole camera model (see Figure 3-1), Gustavsson described a new mode with a compensate term in radial direction (*Gustavsson,* 2000):

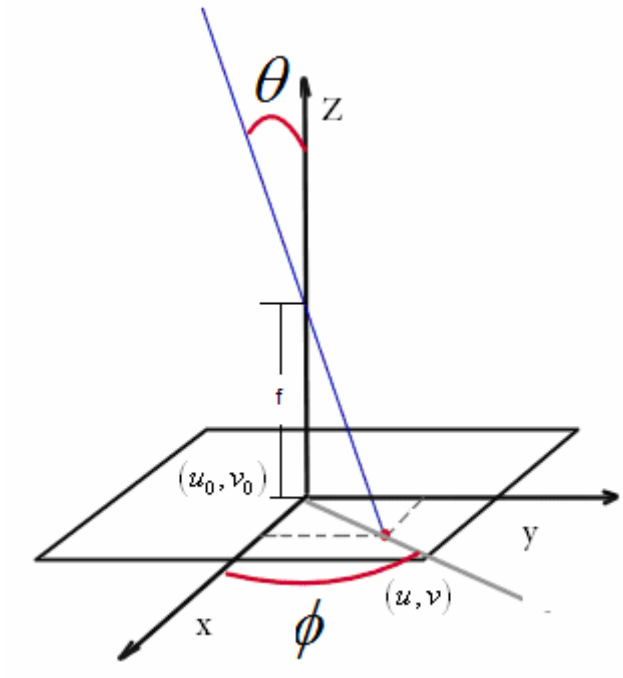


Figure 3-1 Pinhole Camera model

$$\begin{aligned} u &= f_u\left(\left(1-\alpha\right)\cdot\tan\left(\theta\right)+\alpha\cdot\theta\right)\cdot\sin\left(\phi\right)+u_0 \\ v &= f_v\left(\left(1-\alpha\right)\cdot\tan\left(\theta\right)+\alpha\cdot\theta\right)\cdot\cos\left(\phi\right)+v_0 \end{aligned} \tag{3.3}$$

Where:

u : Horizontal coordinates in the image plane,

v : Vertical coordinates in the image plane,

f_u : Horizontal width,

f_v : Vertical width,

u_0 : Horizontal projection of optical axis,

v_0 : Vertical projection of optical axis,

θ : Polar angle relate to optical axis.

ϕ : Azimuth angle around optical axis.

Star positions selected in the Yale Brightness Star catalogue (YBS) and identified star positions found in image are used as interest points to calculate optical parameters by using optimization tools.

3.2.2. Pixel Field-of-view and effective area

With knowledge of optical parameters, the pixel field-of-view (see Figure 3-2) is calculated as (*Gustavsson*, 2000):

$$d\Omega(u,v) = \sin\theta d\phi d\theta = \sin\theta \left|\frac{\partial(\phi,\theta)}{\partial(u,v)}\right| dudv = \sin\theta \left|\frac{\partial(u,v)}{\partial(\phi,\theta)}\right|^{-1} dudv \qquad (3.4)$$

Where, Ω is Pixel field-of-view.

$\left|\frac{\partial(\phi,\theta)}{\partial(u,v)}\right|$: Absolute value of the determinant of the Jacobian of the optical transfer function.

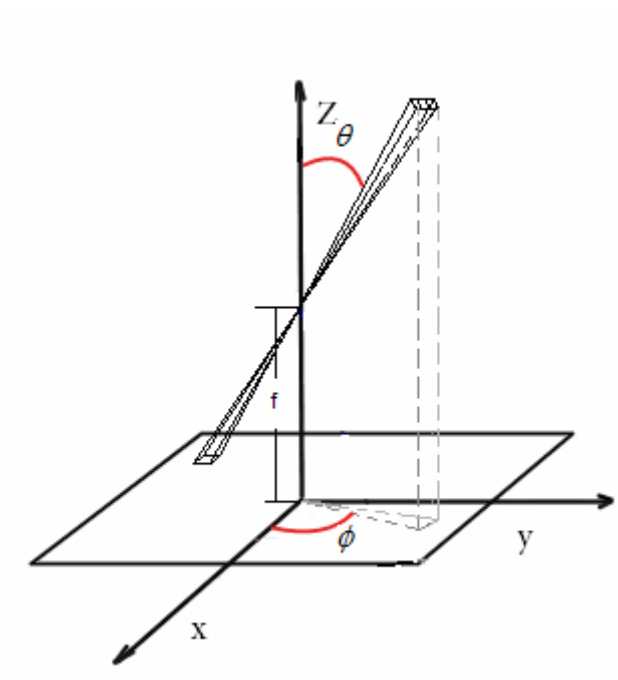


Figure 3-2 Pixel field-of-view

The variation of the effective area with angle θ relative to the optical axis is:

$$A(\theta) = A_L \cos(\theta) \qquad (3.5)$$

Here A_L is the maximum area of the lens of optics, θ is polar angle.

3.3. Results and Error evaluation

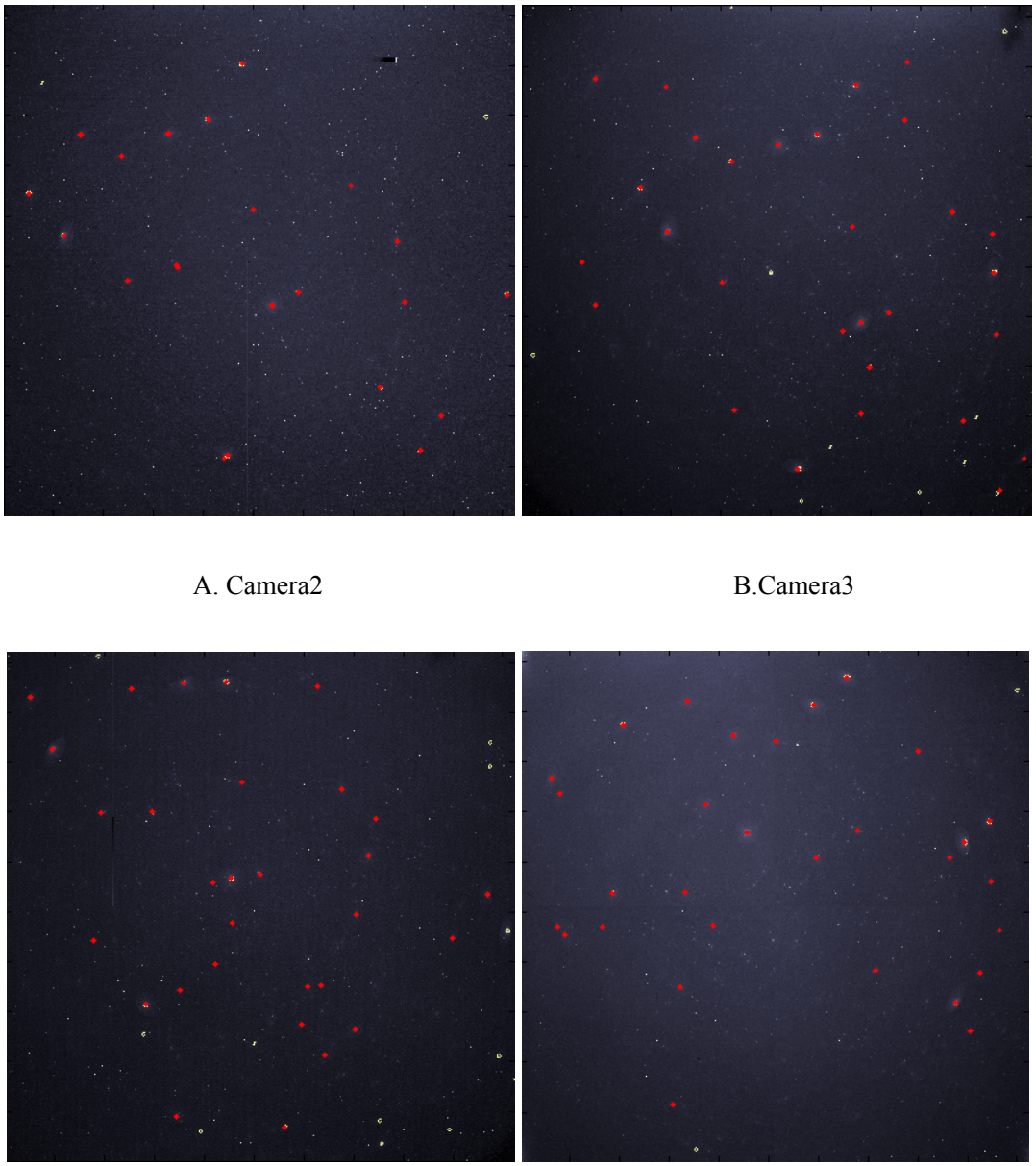
For practical operation, to determine optical parameters, it is necessary to select clear night sky images (in red line, least absorption in atmosphere) with a large number of stars in an image -- preferably more than 20 homogenous distributed in the image plane. To fulfill the requirement that the accuracy of line-of –sight be within 0.02°, the sky position of stars used for the interest point should be rigorously corrected for precession when calculated from the star catalogue (*Gustavsson*, 2000).

A set of tools, provided by Gustavsson, can be found at http:// www.irf.se/~Björn/starcal/. See Table 3-1

Package	Description
Starcal	Program for determination of camera rotations and camera characteristics. Geometrical camera calibration is of vital importance for more advanced image analysis such as tomographic inversion, stereoscopic triangulation. For existing methods it is necessary to accurately know the pixel line-of-sight of each and every image pixel.
Camera	There are a set of optical transfer functions supported (and their inverses). The optical transfer functions determine where in the image light coming from a direction (phi, theta) will project. The inverse of the transfer functions determine the direction of the field-of-view of a pixel (x, y).
Fits_tools	Functions for reading and writing fits files, and related image correction functions.

Table 3-1 Summary of tool kits used for geometrical calibration

Figure 3-3 shows identified stars plotted in a night sky image (red dots) and projected positions from a star catalogue (yellow circles) for each camera. Optical parameters are calculated by using an optimization method to minimize the error between the positions of stars in the image and star positions from a sky map.



A. Camera2 B.Camera3

C. Camera4 D.Camera5

Figure 3-3 Identified stars in camera field-of-view, because technical problems for one quadrant of camera 1, and no laboratory night sky data available for camera 6, only optical parameters of camera 2-camera 5 are calculated. The night sky data are taken in Aug 4th, 2008, 9:33pm.

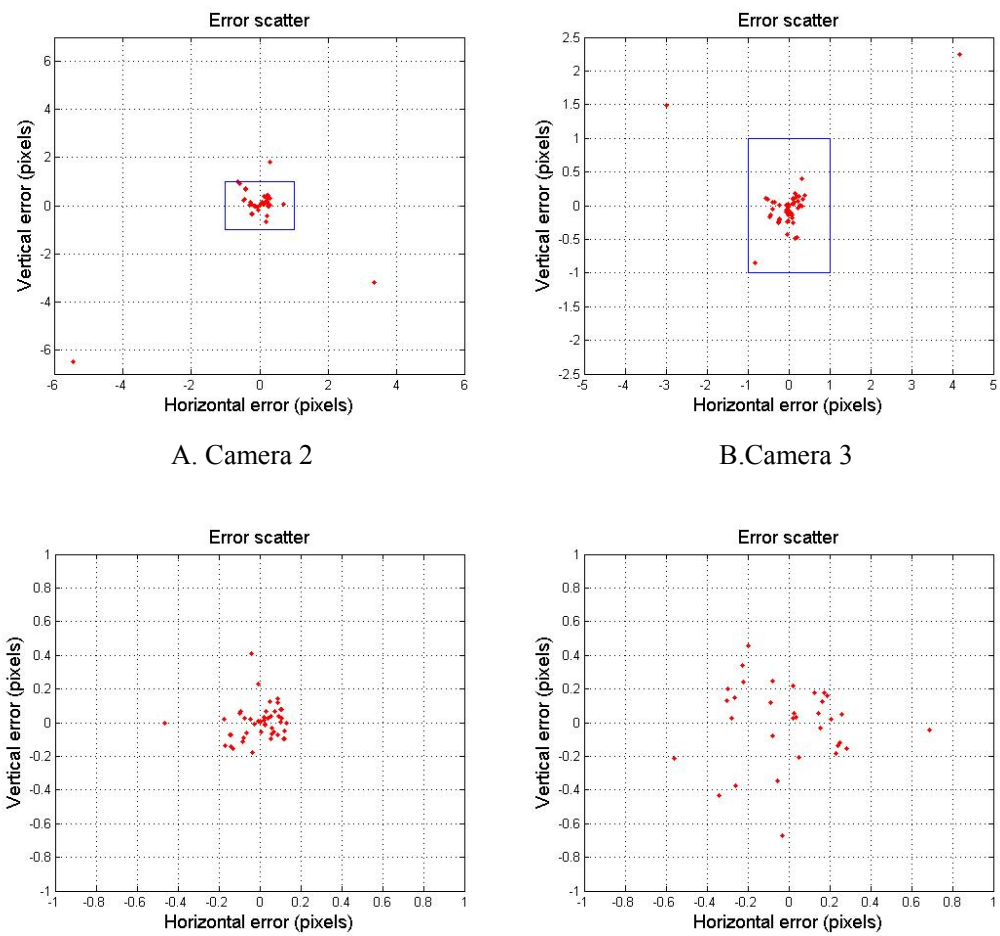
Figure 3-4 shows the two dimensional error between the star positions from a night sky image and the projection of the star catalogue, which are represented by formula(3.6). And Table 3-2 summarized the optical parameters for 5 ALIS imagers.

$$error = \left(\left(u' - \tilde{u} \right)^2 + \left(v' - \tilde{v} \right)^2 \right)^{\frac{1}{2}} \tag{3.6}$$

Here $\left(\tilde{u}, \tilde{v}\right)$ are projection positions from the star catalogue and $\left(u', v'\right)$ are the identified image positions of stars (*Gustavsson*, 2000). Optimization methodology is used to find the minimum value of

$$\sum_{i=1}^{N} \left(\left(u_i' - \tilde{u}_i \right)^2 + \left(v_i' - \tilde{v}_i \right)^2 \right)^{\frac{1}{2}} \tag{3.7}$$

Where N is the number of stars identified in an image. It can be clearly seen that the error spread is essentially confined to one pixel width.



A. Camera 2

B.Camera 3

C.Camera 4

D.Camera 5

Figure 3-4 Error scatter plot

Camera	f_u	f_v	alpha	Beta	gamma	dx	dy	Alfa
2	-0.9945	1.0094	-26.680	-7.6313	-5.3789	-0.0042	-0.0507	-0.1788
3	-1.0465	1.0731	-10.691	9.2450	-1.8353	0.1594	-0.1588	1.1761
4	-1.0141	1.0141	0.1776	0.8296	2.8302	-0.0295	0.0499	0.3575
5	-1.0225	1.0233	21.7662	35.3067	0.6201	-0.0211	0.0362	0.3723
6	-0.5772	0.5749	-9.94e-4	-2.2607	0.7766	-0.0288	0.0089	-0.0163

Table 3-2 Optical parameters for 5 ALIS imagers

Chapter 4

Absolute calibrating ALIS

4.1. Introduction

This chapter presents detailed information in ALIS image pre-processing and absolute calibration with non-linearity correction to find the pixel sensitivity value $\frac{[Rayleighs]}{[Count]}$.

The calibration techniques used for ALIS imagers are similar with the procedure used to calibrate a photometer. However, for a photometer, there is only one detector, but for ALIS imagers, there are around 10^6 detectors (pixels) covering a field-of-view of about $50^\circ \sim 90^\circ$. Thus, calibrating imagers is much more difficult compared with calibrating a photometer. Procedures of absolute calibrate ALIS imagers was described by Brändström and a round of absolute calibration was carried out in 1997(*Brändström,* 2003) under the assumption that all imagers with good linearity at each operational modes. However, by checking laboratory data from Sept. 2007, it has been found that imagers appear non-linearity as pixel intensity growing over a certain counts number. Thus, a new round of absolute calibration for ALIS imagers is strongly desired and a new absolute calibration coefficient table should be used. Due to this non-linearity has been found, a non-linearity correction is proposed before ALIS absolute calibration, in order to obtain a more accurate pixel sensitivity value. This chapter presents detailed procedures regarding this latest ALIS calibration.

4.2. Removing CCD signature

Like astronomical CCD image processing (For example, *Massey*, 1992, *Howell,* 1992 and *West*), in order to absolute calibrate ALIS CCDs, data reduction should be carried out firstly (*Brändström*, 2003).

The process of removing the CCD signature consists of removing the bias, thermal contribution, non-linearity correction and dividing the resultant image by the flat field in order to standardize the response of each image pixel.

$$Signature_removed = f\left(Raw - Bias - Overscan\right)/Flat \tag{4.1}$$

The bias frame, over-scan-strip and nonlinearity should be removed firstly from the raw data, and then divide the data by flat field. See function(4.1), where $f(\cdot)$ is the non-linearity correction function.

4.2.1. Bias Frame

A bias frame is essentially an offset image obtained from an image sensor, with zero integration time. Bias frame appears different according to different camera, bin and gain (see Figure 4-3), thus for ALIS imagers, there are 54 bias frames in total (6 imagers$\times$3gains$\times$3binning).

The obtained bias frame image only contains noise from the amplifier and electronics when readout the sensor data, without any noise from charge accumulation (e.g. from dark current). On an average, a series of bias images, usually more than 50, are used to reduce this noise in our ALIS bias frame calculation. Because of bias changes over time, it would severely impair temporal resolution if a large number of zero exposures were taken in regular intervals. Most CCDs are equipped with some extra pixels at the edge of each line. See Figure 2-2. Those pixels, shaded from light, are called bias pixels, over-scan strips or reference pixels.

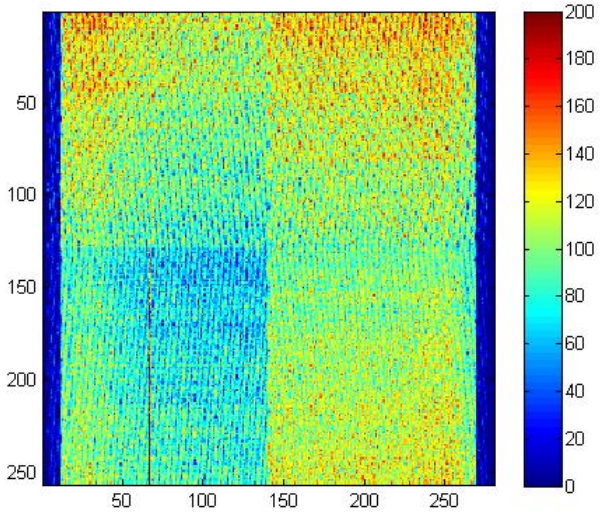
In bias frame, the bias can be expressed as follows (*Brändström*, 2003):

$$B_{i,j} = B_{Overj} + B_{\bar{z}i,j} + B_0 + B_q\left[counts\right] \tag{4.2}$$

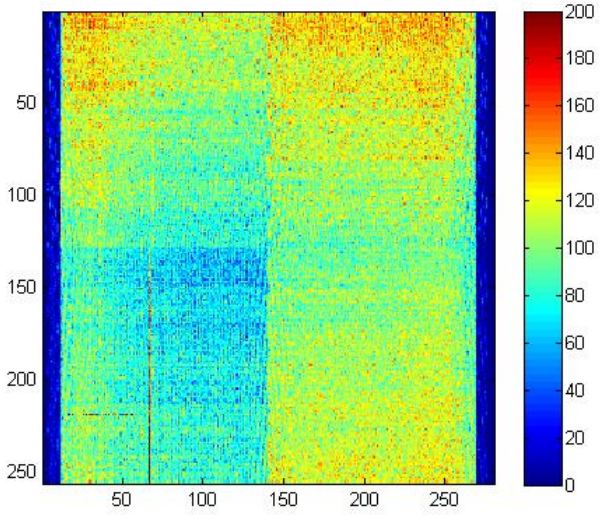
Where,

B_q : Bias variations between the four quadrants.

$B_{i,j}$: is the digital output of the bias frame, the indices are pixel columns, i, and pixel rows, j. Sometimes the images are suffered from interference (see Figure 4-1). It is the result of constructive and destructive interference of electronics.



A. Bias frame with interference



B. Bias after removing interference

Figure 4-1 Removing fringes (Camera4, Gain 32, Bin 4×4)

Such interference patterns can be removed by simply using Fast Fourier Transform and a bandwidth filter for each line in each quadrant.

B_0 : is the preset black level, as given in Table 4-1. This value is set in the CCD configuration file.

CCD	Preset bias B_0
1	-25000
2	-30758
3-4	-30768
5-6	-31768

Table 4-1 Black level for ALIS imagers (*Brändström*, 2003)

B_{Overj} : is the reference pixels value, which is usually obtained by averaging pixels selected in middle columns of over-scan strips on each line (see Figure 4-2, not all are usable).

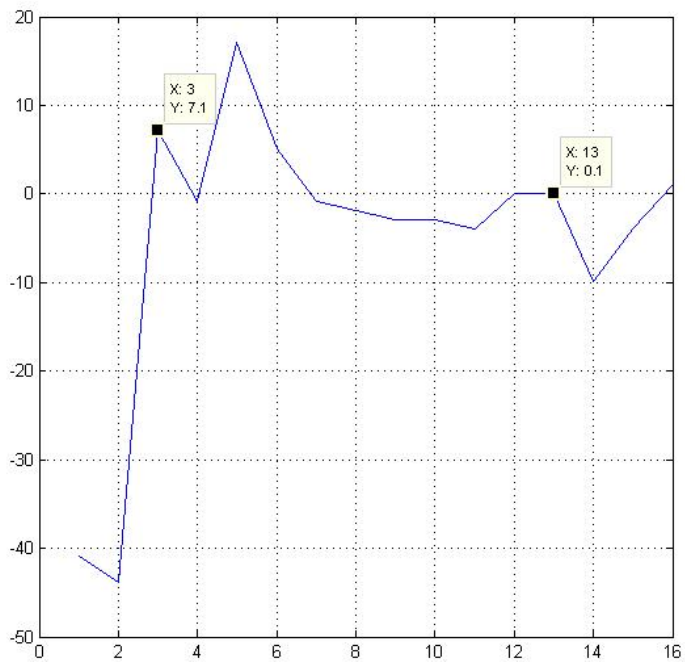
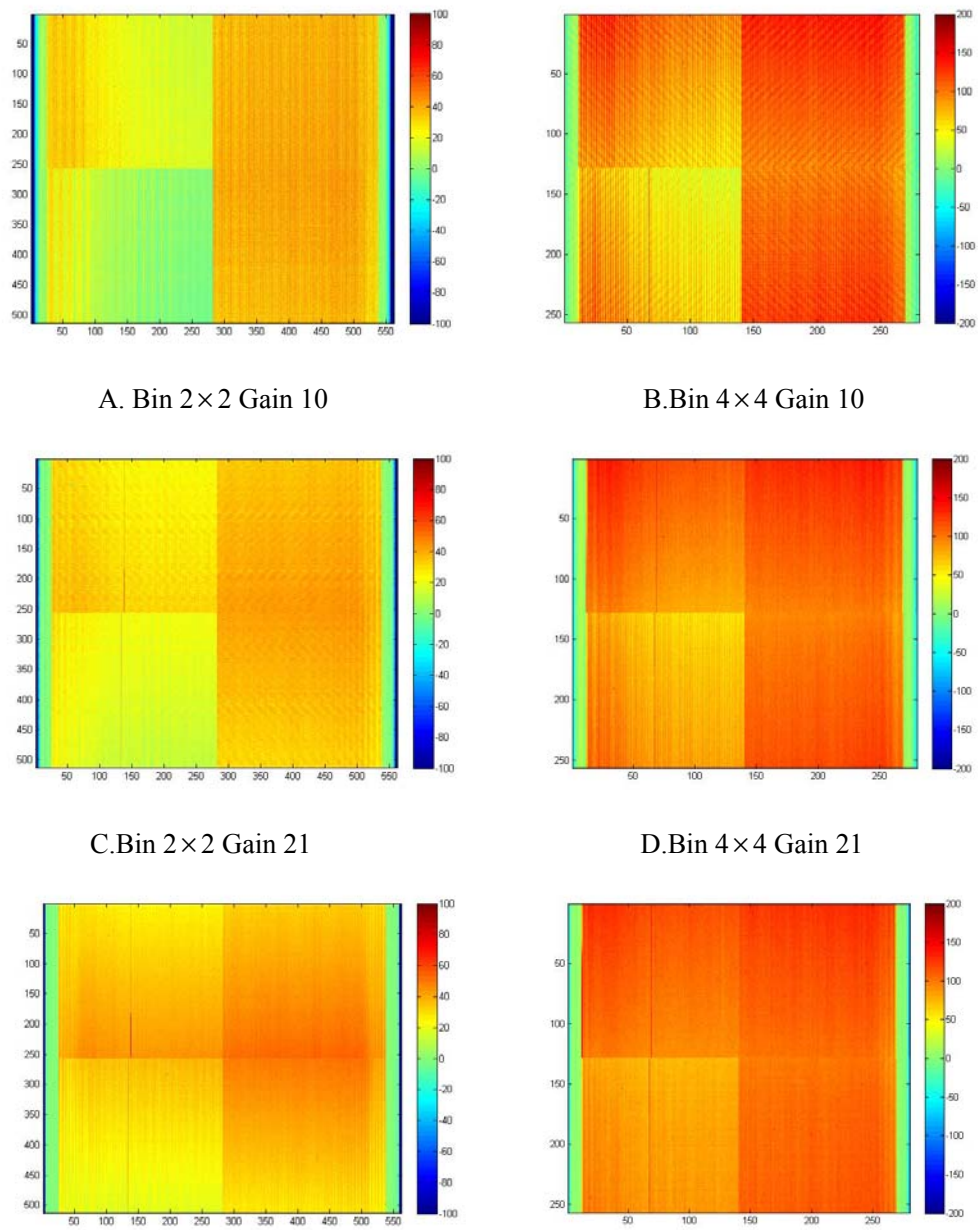


Figure 4-2 Over-scan strips plots for Camera 5, 2007090416374000C.fits, at row 20, column 1 to 16 , to avoid dramatic variation of pixels value at two sides of over-scan strips, averaging values are calculated by using column 3-13.

As mentioned before, to reduce read out noise, it is possible to average 50 bias-frames to get the bias-correction image, $B_{\bar{z}i,j}$:

$$B_{\overline{zi},j} = \frac{1}{N}\sum_{k=1}^{N}\left(B_{i,j} - B_{overj} - B_q - B_0\right) \tag{4.3}$$

During the calibration procedures, discontinuous could be removed by subtracting those bias-correction images.



A. Bin 2×2 Gain 10

B.Bin 4×4 Gain 10

C.Bin 2×2 Gain 21

D.Bin 4×4 Gain 21

E.Bin 2×2 Gain 32

F. Bin 4×4 Gain 32

Figure 4-3 Bias frame of camera 4, even for the same camera, in a different case (different Bin, Gain and Quadrant), the bias frame is totally different and no regular pattern exist.

4.2.2. Dark Frame

A dark frame (an image of noise caused by dark current electrons in an imager) is an image captured with long exposures ($t_{\text{int}} \geq 600s$ for ALIS imagers) with shutter closed. Noise generated by the "leak" electrons due to dark current, spurious charge and residual charge can be shown in a dark frame. Dark frame subtraction has been done for some time in scientific imaging to eliminate this fixed-pattern noise.

For ALIS imagers, the correction has been found to be negligible for integration times less than 10-15 minutes. Therefore, no dark current correction is needed, except for special cases (*Brändström*, 2003).

4.2.3. Non-Linearity correction

For practical work, the linearity is checked by varying the integration time and the CCDs are illuminated by same light standard laboratory source. Usually, for laboratory data, there are four different exposure times for each operational mode (different bin and filter, see Table 4-2).

Filter	Exposure time (seconds)											
Bin mode	Bin 1×1				Bin 2×2				Bin 4×4			
Red (6300 Å)	10	20	30	40	2	5	10	20	2	3	4	5
Green (5577 Å)	10	20	30	40	2	5	10	20	2	3	4	5
Blue (4278 Å)	30	60	120	200	10	20	30	60	2.5	5	10	15

Table 4-2 Exposure times for each operational mode and filter

ALIS cameras are equipped with shutters that open and close rapidly, in the order of magnitude of 10ms (*Brändström*, 2003). For laboratory data taken from Sept. 2007, exposure times varying from 2s to 200s, the relative error in pixel exposures changes from 0.5% to 0.005%. Usually we consider the accuracy is good enough if exposure times lager than 2s. For ALIS imagers linearity testing, images identified by 'Filt'=5, illuminated by source IRF, UJO 920B (4278 Å), are selected for a linearity test because the error caused by shutter characteristics can be ignored by its longest exposure times $\left(t_{\text{int}} > 2.5s\right)$.

Linearity has been checked for 5 imagers at different gain and bin modes. It is impossible list the entire test results here (there are 5imagers$\times$3gains$\times$3binning=45 in total), but a brief description is list in Table 4-3. As an explanation, testing results from the camera 2 and a camera 5 are indicated in this chapter. (See Figure 4-4) which under the operational mode for these results is the most common one used for atmospheric observation, gain 20, bin 4 $\times$ 4.

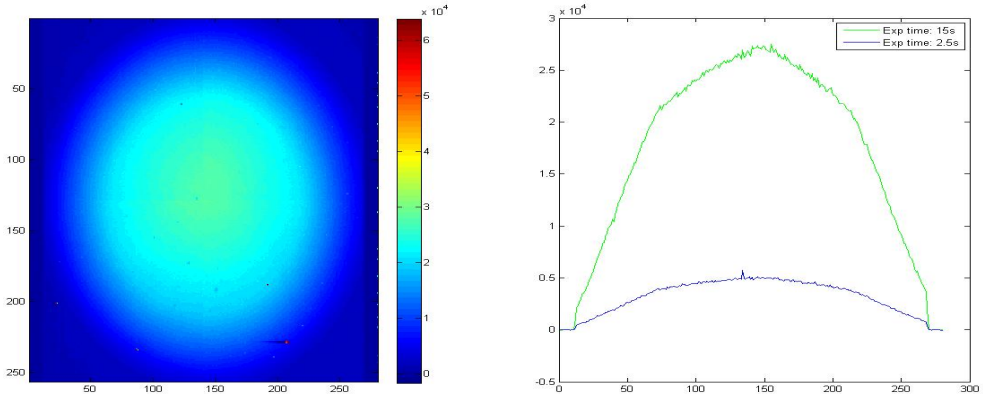
Camera	Description
2	Linearity is not good enough for all operational modes
3	Not good Linearity for all operational modes
4	Non-Linearity exist for low gain mode, and for modes bin 2×2 and 4×4
5	Non-Linearity exist for modes bin 2×2 and 4×4
6	Non-Linearity exist when bin 2×2 and 4×4

Table 4-3 Brief description for 5 imagers linearity

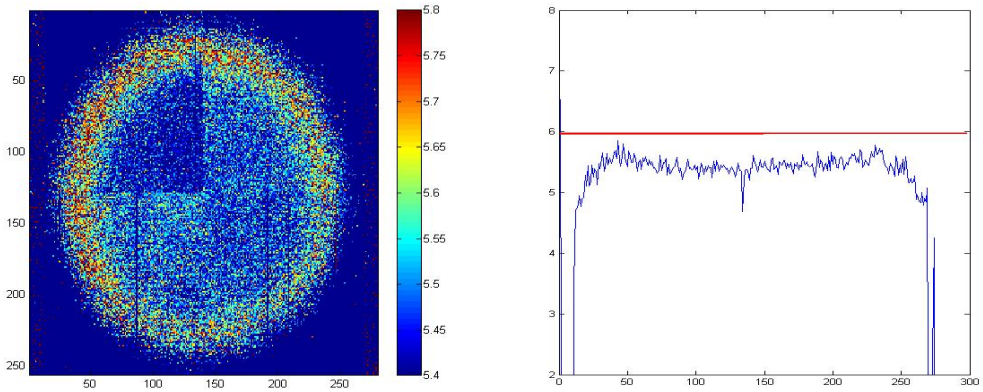
For a camera with complete linearity, the ideal ratio of counts from same pixel between two images should equal the ratio of exposure time. For example, a set of images taken by same camera at gain 20, bin 4×4, with different exposure time 2.5s, 5s, 10s, 15s, the ideal counts ratio should be. (See Table 4-4)

Exposure times	2.5s	5s	10s	15s
2.5s	1	2	4	6
5s	0.5	1	2	3
10s	0.25	0.5	1	1.5
15s	0.16666	0.33333	0.66666	1

Table 4-4 Ideal ratio between images' counts with two exposure times

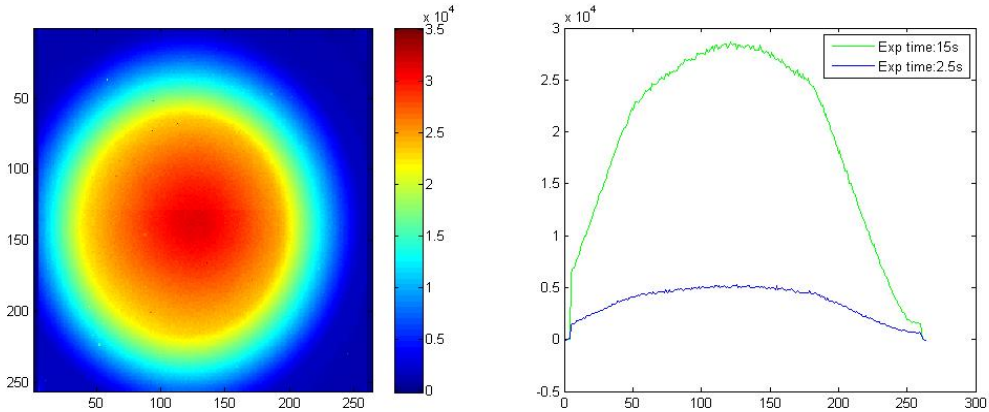


A. Raw data for exposure time 15s B. Line plot of raw image at exposure time 2.5s and 15s

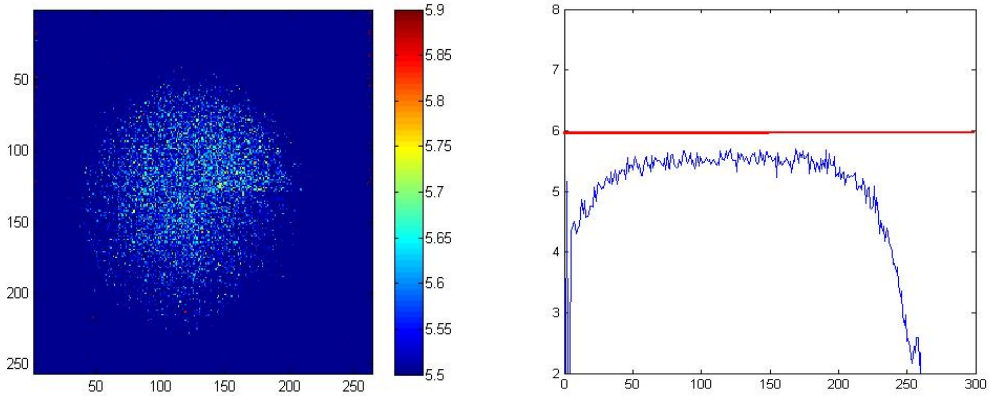


C. Ratio map D. Line plot of ratio map at row 100

Figure 4-4 The linearity checking by using exposure time 15s and 2.5s images with IRF, UJO 920B source for camera 2. The ratio map of camera 2 with average value 5.6035 (for central pixels), error between with idea ratio is 6.6%. Laboratory calibration data are collected from Sept. 2007.



A. Raw data for exposure time 15s B. Line plot of raw image at exposure times 2.5s and 15s



C. Ratio map D. Line plot of ratio map at row 100

Figure 4-5 The linearity checking by using exposure times 15s and 2.5s images with IRF, UJO 920B source for camera 5. The ratio map of camera 5 with average value 5.6130 (for central pixels), error between with idea ratio is 6.4%. Laboratory calibration data are collected from Sept. 2007.

However, it is clear to see from Figure 4-4 and Figure 4-5, for the camera 2, linearity of the central pixels is worse, caused by its higher intensity than those at the boundary, and for the camera 5, all pixels' digital value are around 6.4% below the ideal value. Thus a non-linearity correction method is desired to improve the linearity of 6 ALIS imagers.

A correction function should fulfill the following requirements:

$$\begin{aligned} DN'_{(\text{int_1})i,j} &= f\left(DN_{(\text{int_1})i,j}\right) \\ DN'_{(\text{int_2})i,j} &= f\left(DN_{(\text{int_2})i,j}\right) \\ DN'_{(\text{int_3})i,j} &= f\left(DN_{(\text{int_3})i,j}\right) \\ DN'_{(\text{int_4})i,j} &= f\left(DN_{(\text{int_4})i,j}\right) \end{aligned} \tag{4.4}$$

With constraints,

$$\frac{DN'_{(\text{int_4})i,j}}{DN'_{(\text{int_1})i,j}} = \frac{\text{int_4}}{\text{int_1}}, \frac{DN'_{(\text{int_3})i,j}}{DN'_{(\text{int_1})i,j}} = \frac{\text{int_3}}{\text{int_1}}$$
$$\frac{DN'_{(\text{int_2})i,j}}{DN'_{(\text{int_1})i,j}} = \frac{\text{int_2}}{\text{int_1}}, \frac{DN'_{(\text{int_3})i,j}}{DN'_{(\text{int_2})i,j}} = \frac{\text{int_3}}{\text{int_2}} \tag{4.5}$$
$$\frac{DN'_{(\text{int_4})i,j}}{DN'_{(\text{int_2})i,j}} = \frac{\text{int_4}}{\text{int_2}}, \frac{DN'_{(\text{int_4})i,j}}{DN'_{(\text{int_3})i,j}} = \frac{\text{int_4}}{\text{int_3}}$$

Where,

$DN'_{(\text{int_}k)i,j}$: Corrected digital counts for pixel in image coordinates at i row, j column with integration times (exposure times) k.

$DN_{(\text{int_}k)i,j}$: Raw digital counts for pixel in image coordinates at i row, j column with integration times (exposure times) k.

After a set of functions were tested and evaluated by using laboratory data, we found that quadratic and cubic functions are the most reasonable in the following form. And it should be unique and independent to the exposure times,

$$DN'_{(\text{int_1})i,j} = DN_{(\text{int_1})i,j} \cdot \left(1 + \left(b \cdot DN_{(\text{int_1})i,j} + c\right)\right)$$
$$DN'_{(\text{int_2})i,j} = DN_{(\text{int_2})i,j} \cdot \left(1 + \left(b \cdot DN_{(\text{int_2})i,j} + c\right)\right)$$
$$DN'_{(\text{int_3})i,j} = DN_{(\text{int_3})i,j} \cdot \left(1 + \left(b \cdot DN_{(\text{int_3})i,j} + c\right)\right) \tag{4.6}$$
$$DN'_{(\text{int_4})i,j} = DN_{(\text{int_4})i,j} \cdot \left(1 + \left(b \cdot DN_{(\text{int_4})i,j} + c\right)\right)$$

Or,

$$DN'_{(\text{int_1})i,j} = DN_{(\text{int_1})i,j} \cdot \left(1 + \left(a \cdot DN^2_{(\text{int_1})i,j} + b \cdot DN_{(\text{int_1})i,j} + c\right)\right)$$
$$DN'_{(\text{int_2})i,j} = DN_{(\text{int_2})i,j} \cdot \left(1 + \left(a \cdot DN^2_{(\text{int_2})i,j} + b \cdot DN_{(\text{int_2})i,j} + c\right)\right)$$
$$DN'_{(\text{int_3})i,j} = DN_{(\text{int_3})i,j} \cdot \left(1 + \left(a \cdot DN^2_{(\text{int_3})i,j} + b \cdot DN_{(\text{int_3})i,j} + c\right)\right) \tag{4.7}$$
$$DN'_{(\text{int_4})i,j} = DN_{(\text{int_4})i,j} \cdot \left(1 + \left(a \cdot DN^2_{(\text{int_4})i,j} + b \cdot DN_{(\text{int_4})i,j} + c\right)\right)$$

Because, for images with very short exposure times, $DN_{(\text{int_1})i,j} \to 0$, these kinds of equations approach linearity under the condition that parameters a, b, c are very small

values (in the order of magnitude 10^{-4}) which close to zero. Along with extended exposure time, $DN_{(\text{int_1})i,j} \to \infty$, those equations start to show non-linear characteristics.

Optimization tools which best minimizes the value of objective function(4.8) are selected to find the a, b, c parameters for the correction function. Note that, the notation $s_{k_1 k_2}$ means square root of sum of squared residuals between real ratio and ideal ratio. With the knowledge a, b, c should be very close to zero, i.e. a minimum solution should exist around the point (0, 0, 0). Thus, a Trust Region algorithm (fminunc function in matlab optimization tool box) is selected for this task because it can converge to the nearest optimized value from an initial point very quickly.

$$s_{k_1 k_2} = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} \left(\frac{DN'_{(\text{int_}k_1)i,j}}{DN'_{(\text{int_}k_2)i,j}} - \frac{\text{int_}k_1}{\text{int_}k_2} \right)^2} \tag{4.8}$$

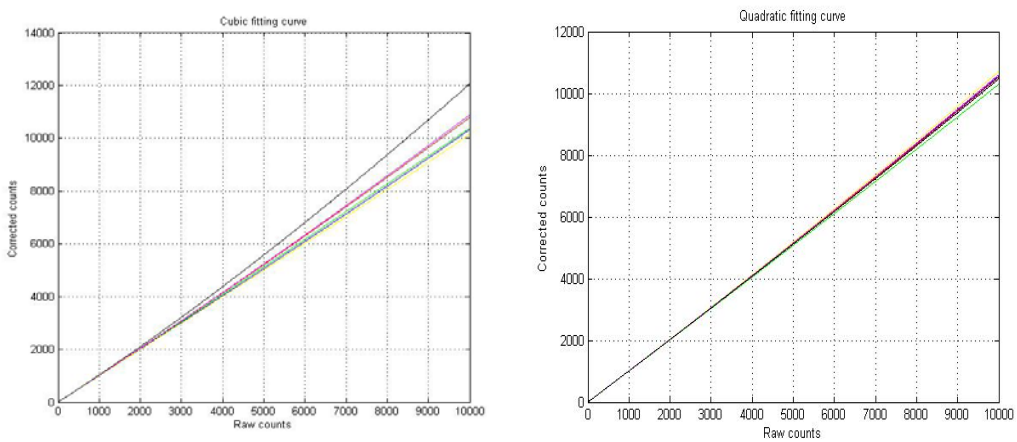
Then, six groups of parameters are calculated by minimizing $s_{k_1 k_2}$, $s_{k_1 k_3}$, $s_{k_1 k_4}$, $s_{k_2 k_3}$, $s_{k_2 k_4}$ and $s_{k_3 k_4}$ respectively.

Calculation results from laboratory data of camera 2, gain 20, bin 4×4, quadrant 0 by using quadratic and cubic function are listed in the Table 4-5. Fitting curves are shown in Figure 4-6. From the results, it is possible to get the following conclusions:

1. By using a cubic function, value of coefficient a is much smaller than b and c. This means the cubic term can be canceled – a quadratic function is good enough for linear correction.
2. The curves of quadratic fitting are overlapped, which is better than cubic fitting, because for cubic function, optimization function has to search a minimum value in 3 dimensional spaces. It is make more difficult to convergence to global optimization point than in 2 dimensions. This means it is much easier to find a unique linear correction function by fitting a quadratic function.

	quadratic			cubic			
	Min	b	c	Min	a	b	c
$s_{k_1k_2}$	2.2253	0.028e-4	2.851e-4	2.2233	-2.4e-10	2.85e-4	1.128e-4
$s_{k_1k_3}$	4.3066	0.055e-4	0.263e-4	4.2657	1.9e-10	1.354e-6	1.186e-4
$s_{k_1k_4}$	6.4236	0.052e-4	0.408e-4	6.2840	-1.3e-10	9.11e-6	8.64e-5
$s_{k_2k_3}$	1.7793	0.070e-4	2.734e-4	1.7320	3.2e-10	-1.80e-6	1.22e-4
$s_{k_2k_4}$	2.5600	0.059e-4	0.393e-4	2.5075	-1.1e-10	9.90e-6	1.50e-4
$s_{k_3k_4}$	1.1225	0.046e-4	1.140e-4	0.9831	-4.7e-10	2.52e-5	-1.24e-5

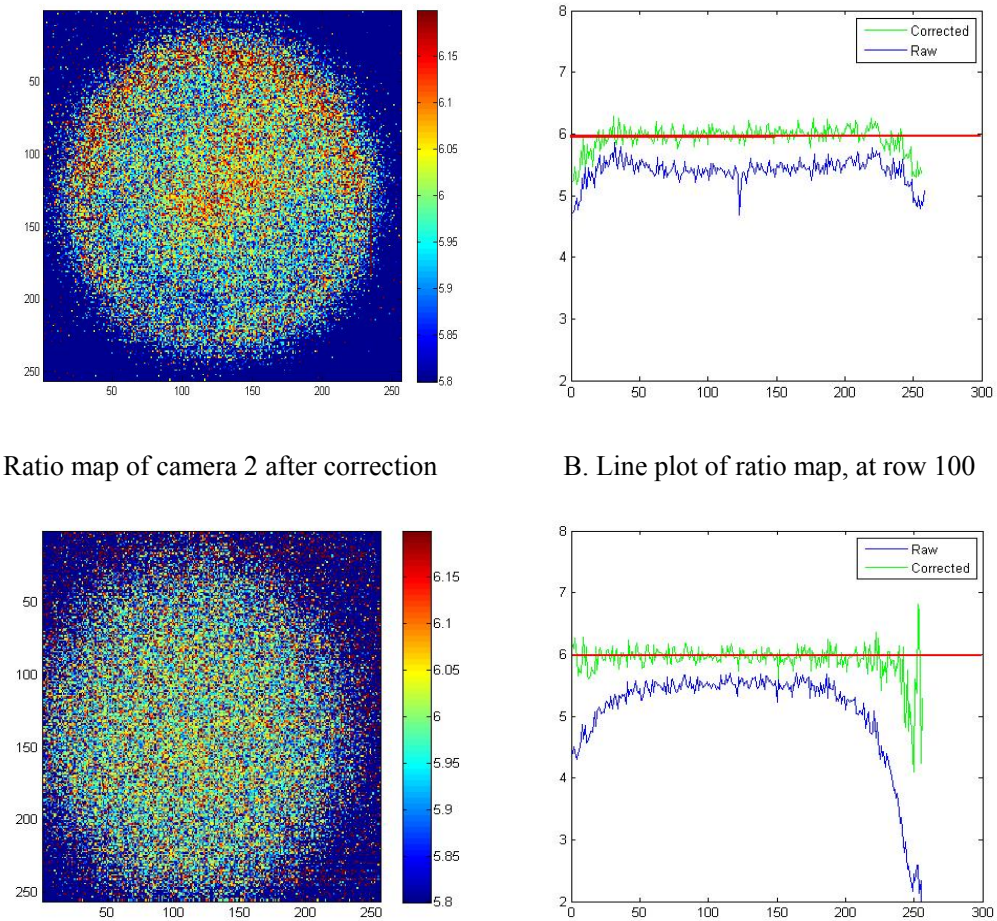
Table 4-5 Parameters are calculated by using quadratic and cubic function



A. Fitiing curves for cubic function B. Fitting curves for quadratic function

Figure 4-6 Curve fitting of non-linearity correction function by using quadratic and cubic function, Legend: curve for $S_{k_1k_2}$ is 'green', $S_{k_1k_2}$ 'red', $S_{k_1k_2}$ 'yellow', $S_{k_1k_2}$,'pink', $S_{k_1k_2}$,'black' and $S_{k_1k_2}$, 'purple'.

One hundred and eighty ($5 imagers \times 4 quadrants \times 3$gain modes$\times 3$bin modes) groups of b and c have been calculated (see Appendix B). Correction results of camera 2 and camera 5(in gain 20, bin 4×4) is shown in Figure 4-7 and Figure 4-8. It can be clearly seen that the error decreases rapidly after non-linearity correction.



A. Ratio map of camera 2 after correction

B. Line plot of ratio map, at row 100

C. Ratio map of camera 5 after correction

D. Line plot of ratio map, at row 100

Figure 4-7 The ratio of counts between exposure times 15s and 2.5s with IRF, UJO 920B source (after linearity correction).A. the corrected ratio map of camera 2 with average value (for central pixels) 6.0034, error between with idea ratio is 0.0006%. B. a line plot of the corrected ratio map at row 100 for camera 2. C. the ratio map of camera 5 with average value 5.9975 (for central pixels), error between with idea ratio is 0.0004%. D. a line plot of the corrected ratio map at row 100 for camera 5.

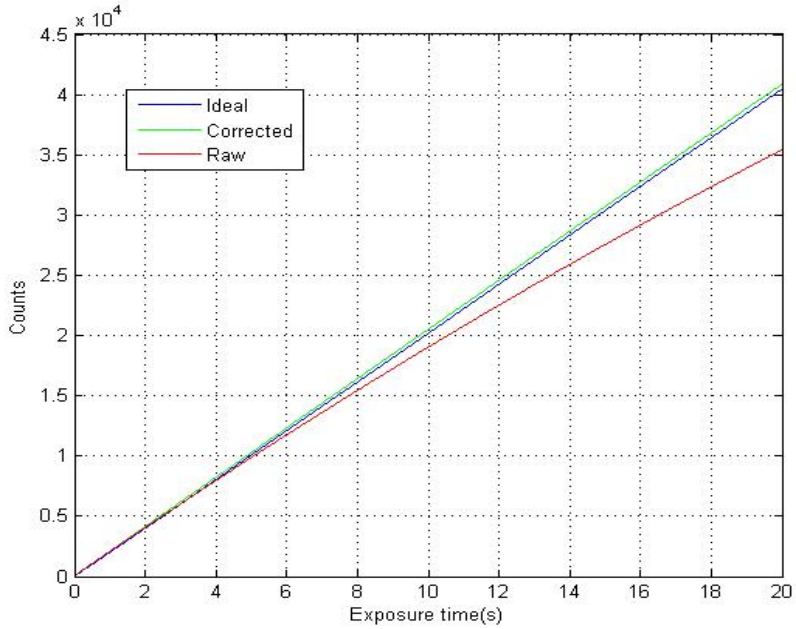


Figure 4-8 Average counts for central pixels against exposure time (Camera 2, Gain 20, Bin 4×4, illuminated by IRF, UJO 920B), for raw data, it appears non-linearity when exposure times longer than 4 seconds, for corrected data (green line), it overlaps with the ideal values and the error less than 0.005%.

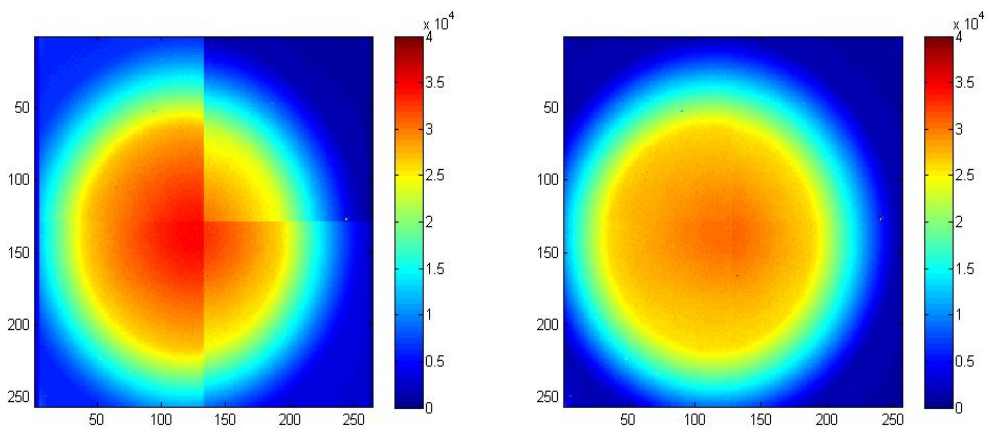
4.2.4. Flat field

The flat field correction is the process to remove the distortion caused by slightly variations of pixel-to-pixel sensitivity to light. These variations are usually introduced by several reasons, different solid angles between pixels, transmittance variations over the filter surface and shadow of dust in optics system, etc. A division by flat field image removes these defects. To obtain a flat field frame from an astronomical imager, a common technique is to use a twilight sky as a uniform light source, and imprint variations, that are caused by pixel-to-pixel sensitivity into an image.

Currently, no suitable flat field correction has been done for ALIS imagers, although several attempts have been tried by using cloudy sky, white screen and diffuse lamp cover, etc. However, none of them obtained a sufficient quality flat field image (*Brändström*, 2003).

Instead, for ALIS imagers calibration, the variation caused by different between solid angle between pixels has been modeled according to the following procedure:

1. Calculate the product value of field-of-view and effective area for each pixel (*Gustavsson*, 2000).
2. Find the maximum product value in the pixel matrix.
3. Normalizing correction matrix by dividing the maximum product value. (flat field matrix)



A. Raw image B. Corrected image by removing CCD signature

Figure 4-9 Raw image (A) and corrected image (B) by removing CCD signature (Camera 5, Gain 13, Bin 4×4, Filter 1, illuminated by source IRF, UJO 920B)

After removing the CCD signature, it appears that the variations caused by pixel-to-pixel sensitivity have been removed in the central part of image, see Figure 4-9.

4.3. Absolute calibration with laboratory data

Inter-calibration is carried out to provide a comparable scale of intensities of aurora or airglow emission. Three radioactive ^{14}C light-standards are inter-calibrated against other light standards during European calibration workshops held at regular intervals. Results of the most resent ^{14}C -sources that are used for ALIS calibration (phosphors: 920B, Lauche lamp and Y275) are summarized in Table 4-6(*Lauche and Widell, 2000, Brändström*, 2003 *and Henricson, 2008*) and the column emission rate is plotted against wavelength in Figure 4-10.

Filter(Å)		3914	4280	4866	5573	5882	6299	6562
Test source								
IRF, Lauche Lamp	2000	0.19	1.1	8.5	50	98	223	454
	2001	0.17	1.2	8	48	115	228	428
	2007	0.15	1.06	8.67	50.75	107.55	225.90	332.82
IRF, UJO Y275	1985	0.03	0.3	3.8	251.0	378	217	113
	1999	0.01	0.2	3.8	263	383	282	165
	2000	0.01	0.22	3.9	276	405	282	181
	2001	0.002	0.18	3.6	258	482	274	155
	2006	0	0.23	3.72	255	398.1	226.6	147.9
	2007	0.15	0.19	3.68	255.69	413.69	260.12	129.8
IRF, UJO 920B	1985	5.1	126	61.5	18.6	10.5	6.7	8.1
	1999	4.4	102	60	22	12	7.6	2.4
	2000	4.6	109	64	23	12.7	10	8.5
	2001	5.2	105	65	22	13	9.2	1.2
	2006	4.9	106.3	61.8	22.8	14.1	9.5	15.9
	2007	4.32	98.23	61.03	22.64	15.21	11.85	4.61

Table 4-6 Results from a selection of recent intercalibration workshop, and the column emission rates are given in [Rayleighs/Ångstrom]

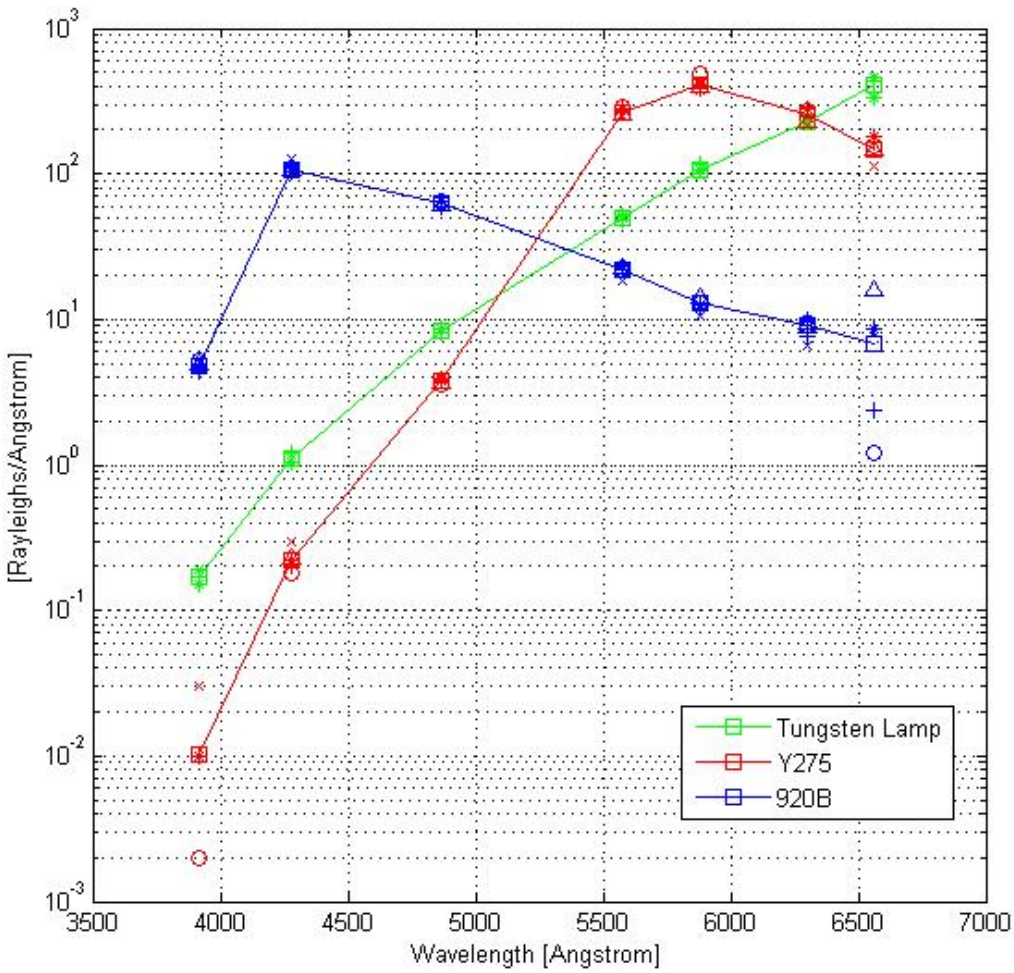


Figure 4-10 Column emission rates [R/Å] against wavelength [Å] for three radioactive ^{14}C light-standards with phosphor: 920B (blue), Lauche Lamp (green), and Y275 (red). Legend: intercalibration sessions of 1981 'x' 1985'+', 1999'*', 2000 'o', 2006'^', and 2007'□'. Lines connect the results from the latest calibration workshop, held in Andoya Rocket Range, Norway, 2007.

Because the light-emitting surface of the light source does not cover the entire field-of-view of the imager, only the pixels located in the central part are used (see Figure 4-11) to find $\overline{DN'}$.

$$C_{abs\lambda c} = \frac{I_{cal} \cdot t_{\text{int}}}{\overline{DN'}} \left[\frac{Rayleighs}{Count} \right] \tag{4.9}$$

Where, $C_{abs\lambda c}$ indicates an absolute calibration constant. The column emission rate, I_{cal}, which is obtained by integral of light standards line (Figure 4-10) over the filter pass band $\Delta\lambda = \lambda_2 - \lambda_1$ (for blue source, 920B, $\Delta\lambda = 50$ Å, for red and green source, Lauche and Y275, $\Delta\lambda = 40$ Å), is given by,

$$I_{cal} = \int_{\lambda 1}^{\lambda 2} I_{ls}(\lambda)\, d\lambda [R] \tag{4.10}$$

And $DN_{i,j}^{'}$ is calculated by,

$$DN_{i,j}^{'} = \frac{f\left(DN_{i,j} - B_{\overline{zi},j} - B_{overj}\right)}{F_{i,j}} [Counts] \tag{4.11}$$

Where $DN_{i,j}$ is raw digit output of pixel at row i, column j, $F_{i,j}$ indicates flat field, and $f(\cdot)$ is non-linearity correction function, equation(4.6).

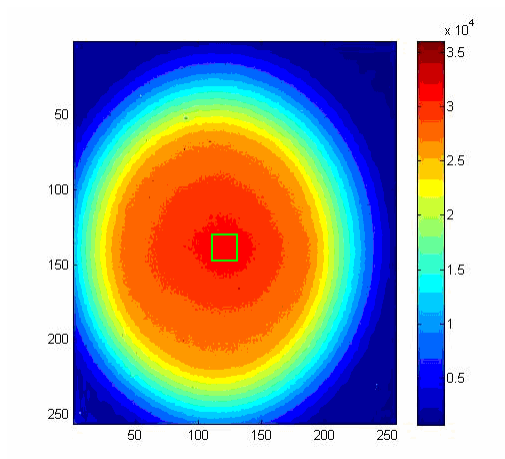


Figure 4-11 Pixels are selected for calibration (in green rectangle)

Absolute calibration should be carried out for each filter, gain and bin of each imager. The results from last absolute calibration of ALIS are included in appendix A.

Chapter 5

Stellar calibration

5.1. Introduction

Visible light emitted from the upper atmosphere or cosmos are mainly absorbed in the troposphere and stratosphere (*Herzburg*, 1965 and *Meier*, 1991). To take this atmospheric absorption and effect of domes, which is made by plexiglass and used for protection camera lens, into account, calibrating ALIS imagers by using stars with well-known spectrum as the calibrator is carried out. In this chapter we present the stellar calibrator samples and the method used for modeling point-spread function (PSF) (*Rufino G, Accardo D*, 2003). For ALIS images, the PSF can be found by fitting the shape of stars in the image planes with long exposure times (*Gustavsson*, 2000). It is well known that the PSF of stars can be fairly well modeled by a two dimensional Gaussian function with different sigma dependent on the location on the image plane. By integral Gaussian fitting, it is possible to calculate the counts produced by stars. The calibration is based on observations that use 15 seconds' exposure times; we demonstrated the coefficient C_{abs} is very stable for those images observed in three hours.

5.2. Collection of digital counts produced by star

To begin with, 404 night sky images were taken by camera 3 and camera 5 on March 14th, 2011, from 9:00 pm to 11:00pm. There are 30 samples for each filter in one hour. Star HR 5191, 5054, 4554, 5563 and 8974 are chosen for ALIS imagers calibration, because they are five of the brightest stars in the night sky images and stellar spectrums information can be found in Pulkovo spectrophotometric catalogue. Figure 5-1 shows a 5 pixel × 5 pixel window, bounded by an inner red rectangle, in one stellar image containing the star scene. This window includes all of the pixels illuminated by the star. And the background can be obtained by fitting the pixel values outside the red window but within the green window using a plane function(5.1).

$$B(x, y) = (a \cdot x + b)(c \cdot y + d) \tag{5.1}$$

After subtracting the background, any pixel value of digital counts within the inner red window is considered as produced by stellar illumination. To calculate the total counts produced by the stellar illumination, it is possible to integrate a Gaussian function (5.2) fitted by using these digital counts.

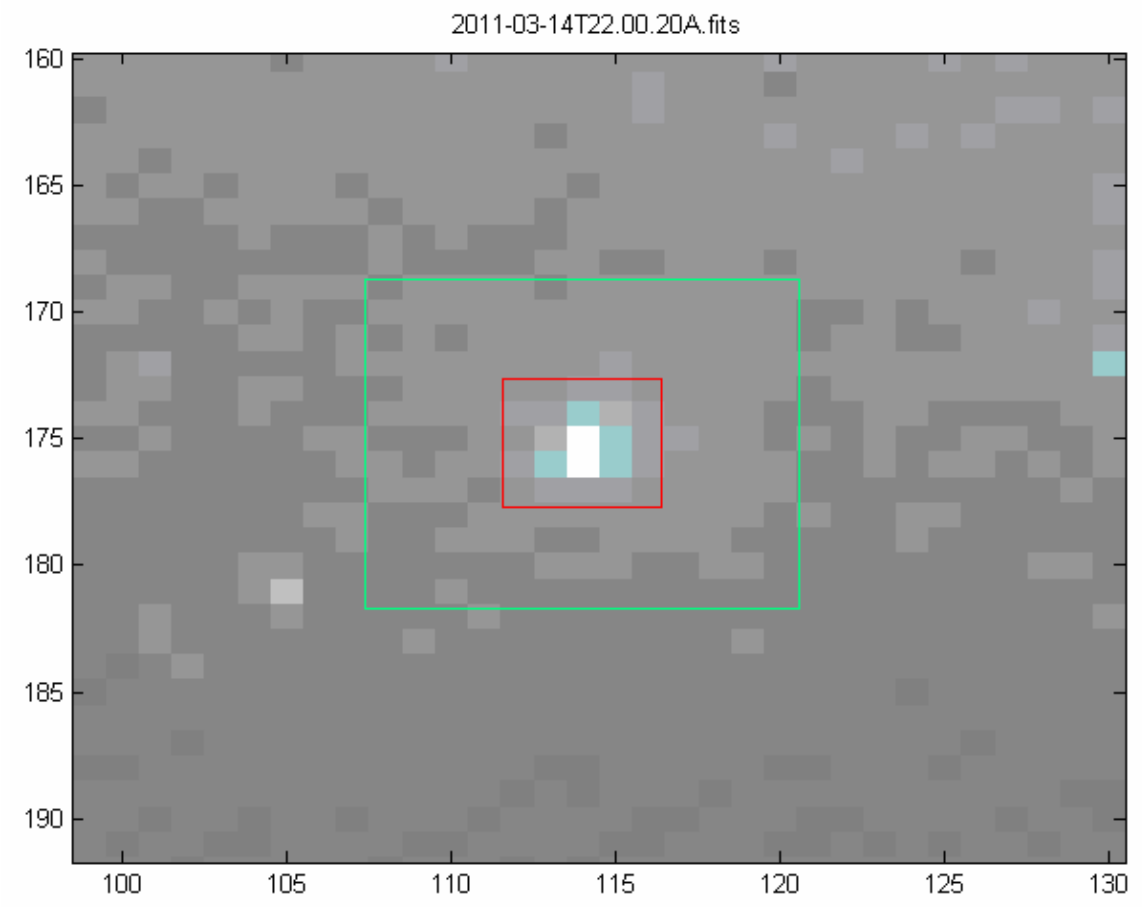


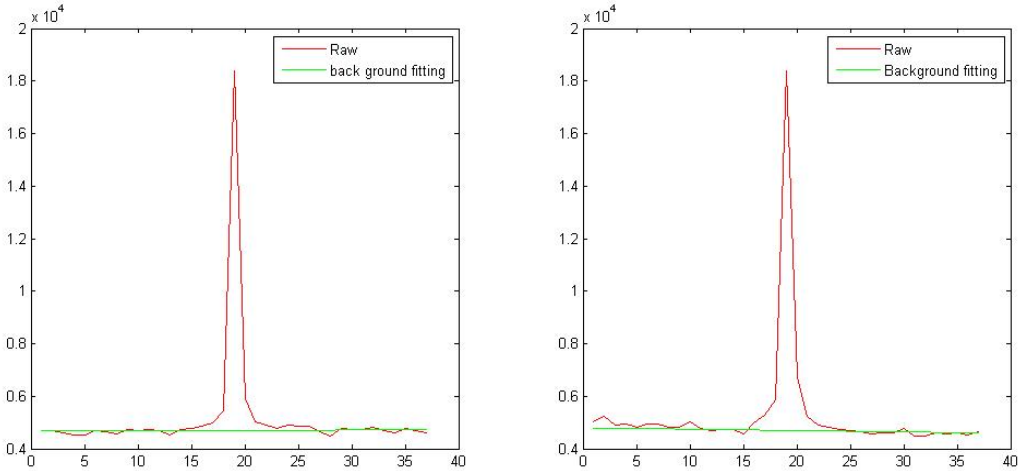
Figure 5-1 Typical star scene in a stellar image, stars located at the boundary of image plane should be excluded as a calibrator, because of the low accuracy of solid angle, the calibrator selected should locate in the central of image plane.

It is well known that stars point spread function is fairly modeled by a two dimensional Gaussian function,

$$f(x,y) = A \cdot \exp\left(-\frac{(x-x_0)^2}{\sigma_x} - \frac{(y-y_0)^2}{\sigma_y} \right) \tag{5.2}$$

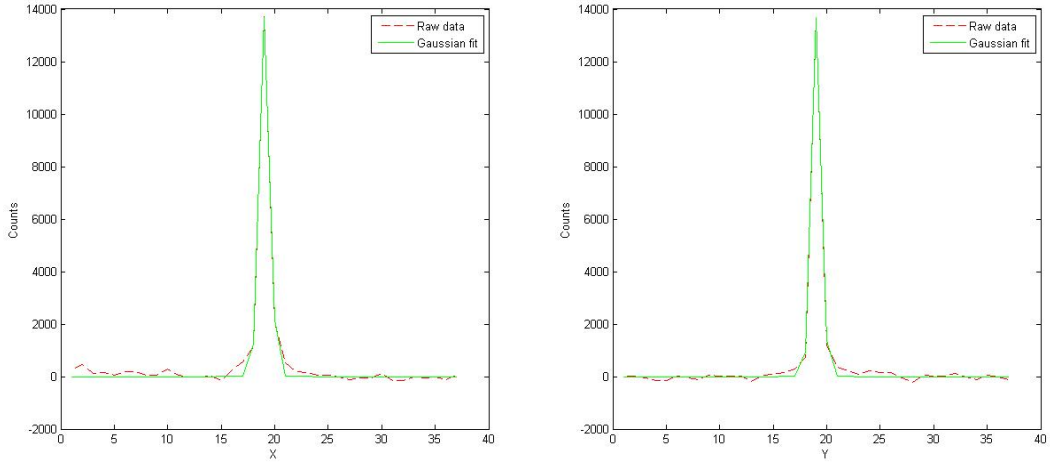
To obtain the best fit, an optimization method is needed which can search minimum values in five dimensional space, $A, x_0, \sigma_x, y_0, \sigma_y$. Several optimization methods exist in matlab optimization tool box to solve this problem.

An example of background fitting is shown in Figure 5-2. And an example of a brightness star profile, together with Gaussian fitting, is shown in Figure 5-3.



A.Line plot at x direction B. Line plot at y direction

Figure 5-2 Background fitting. Sky background can be considered as a slope plane.



A.Line plot at x direction B. Line plot at y direction

Figure 5-3 Gaussian fitting. The dotted curve is the profile of the raw data removed background, and the solid curve shows the Gaussian fit

This fitting process is repeated for 90 images taken by camera 3 and camera 5 for each spectral band. The total digital number of stars is calculated by integrating the Gaussian function,

$$DN = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} A\exp\left(-\frac{(x-x_0)^2}{\sigma_x}-\frac{(y-y_0)^2}{\sigma_y}\right)dxdy = A\pi\sqrt{\sigma_x\sigma_y}\,[Counts] \qquad (5.3)$$

The results are averaged together to obtain the average digital numbers for each star and each band. Table 5-1 and Table 5-2 give the digital numbers of the stellar image evaluations for camera 5 and camera 3 respectively.

V magnitude	HIP	Star name	4278 Å [Counts]	5577 Å[Counts]	6300 Å[Counts]
2.44	4554	Gamma UMa	7.7529e3	1.0480e4	1.4601e4
2.06	5054	Zeta UMa	1.1760e4	1.6474e4	2.3883e4
1.86	5191	Eta UMa	1.4068e4	1.6234e4	2.1714e4

Table 5-1 Averaged digital numbers from stars for each band from ALIS stellar images collected on 14th March, 2011 (Camera 5, gain 20, bin4×4)

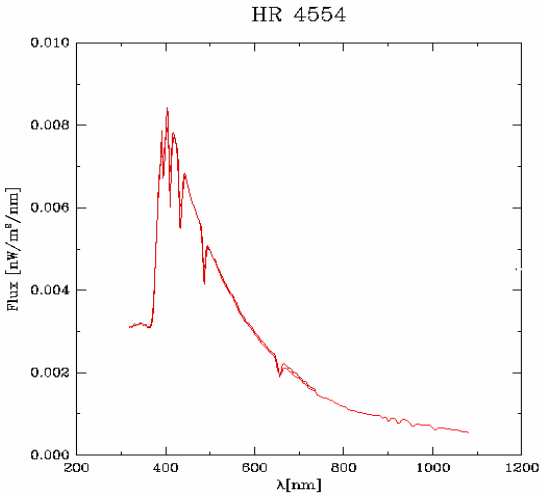
V magnitude	HIP	Star name	4278 Å [Counts]	5577 Å[Counts]	6300 Å[Counts]
2.08	5563	Beta UMi	null	1.8483e4	3.0067e4
3.21	8974	Gamma Cep	null	7.2034e3	1.0984e4

Table 5-2 Averaged digital numbers from stars for each band from ALIS stellar images collected on 14th March, 2011 (Camera 3, gain 20, bin4×4), no results obtained from line 4278 Å due to quite low SNR.

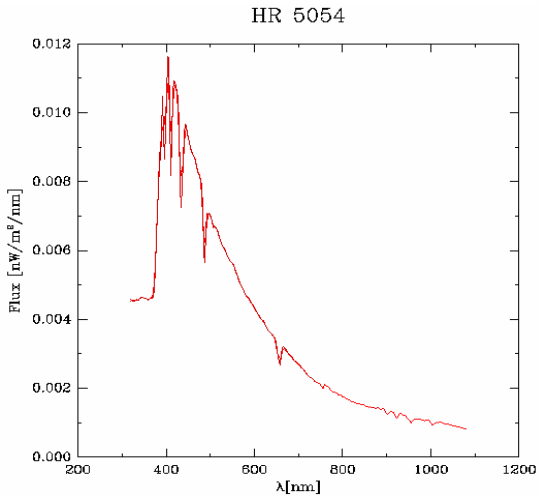
5.3. Absolute calibration with stellar spectrum

Five stars are chosen as calibrators in the Pulkovo spectrophotometric catalogue, which was published in *Baltic Astronomy* Vol. 5, No. 4 (1996). This catalogue combines the results of numerous observations made by Pulkovo astronomers at different observing sites. The catalog consists of three bands of stars: the first contains the data for 602 stars in the spectral range of 320--735 nm with a resolution of 5 nm (1nm=10Å); the second one contains 285 stars in the spectral range of 500--1080 nm with a resolution of 10 nm, and the third one contains 278 stars combined from the preceding catalogs in the spectral range of 320--1080 nm with a resolution of 10 nm. The data are presented in absolute energy units $nW/m^2/nm$, with a step of 2.5 nm and with accuracy not less than 1.5--

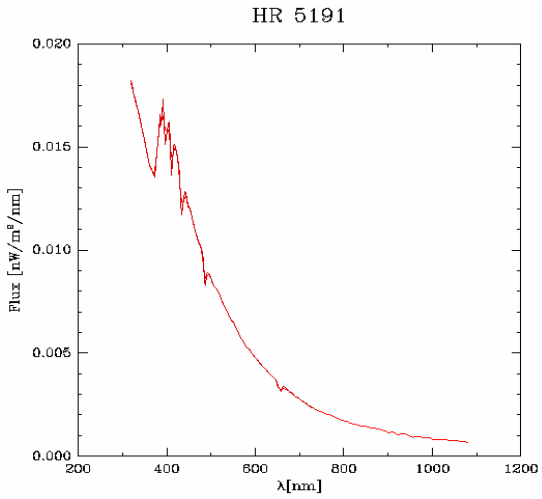
2.0% (*Alekseeva and Arkharov*, 1996). Figure 5-4 shows the stellar spectrum for 5 stars, HR 4554, HR 5054, HR 5591, HR 5563 and HR 8974.



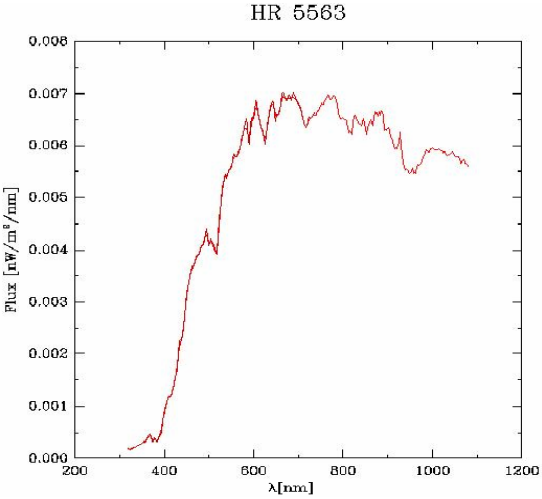
A. Spectrum of Gamma UMa



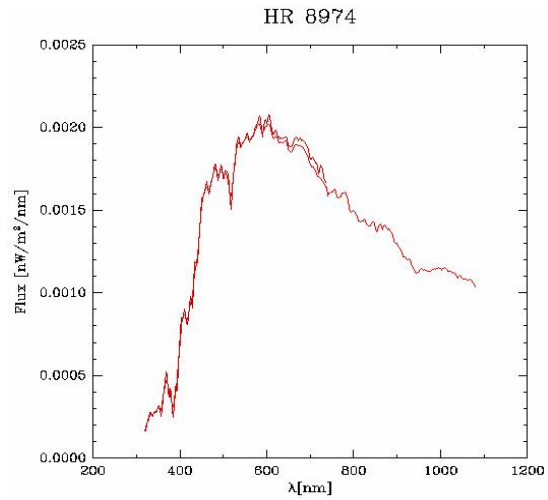
B. Spectrum of Zeta UMa



C. Spectrum of Eta UMa



D. Spectrum of Beta UMi



E. Spectrum of Gamma Cep

Figure 5-4 Stellar spectrum (*Alekseeva and Arkharov*, 1996)

5.3.1. Irradiance to Rayleigh

The unit of stellar spectrum is given in irradiance. In order to compare with calibration results with laboratory data, a unit conversion is needed. *Baker and Romick* (1976) presented the applicability of the Rayleigh as a unit of apparent radiance of extended light sources and proposed his conversion formula,

$$R = 2.103617\pi\lambda L \times 10^{13} \tag{5.4}$$

where, the wavelength λ is expressed in um, and L is the radiance in energy units.

It is also possible to separate the conversion into following steps, for more accuracy:

The spectral irradiance L_E from Pulkovo catalogue is in energy units $\left[\frac{nW}{s \cdot m^2 \cdot nm}\right]$.

It is well-known that energy of a photon is only characterized by its wavelength, denoted by λ. And there is an inverse relationship between the energy of a photon E and its wavelength λ:

$$E = \frac{hc}{\lambda} \tag{5.5}$$

Where h is Planck's constant and c is the speed of light. With the value of h:$6.626 \times 10^{-34}\, J \cdot s$ and c: $2.998 \times 10^{8}\, m/s$.

Thus, it is possible to express the spectral irradiance in quantum units E_r as the following equation(5.6) and Figure 5-5 shows the spectrums for five stars in quantum units:

$$E_\gamma = \frac{L_E \lambda}{hc}\left[\frac{photons}{s \cdot m^2 \cdot nm}\right] \tag{5.6}$$

Dividing by the solid angle of field-of-view for pixels, Ω and multiplying the filter band width $\Delta\lambda$ (for 4278Å is 5nm, for 5577 and 6300 is 4nm).

Then we can obtain the spectral radiance in quantum units,

$$L_\gamma = \frac{\Delta\lambda E_\gamma}{\Omega}\left[\frac{photons}{m^2 \cdot sr \cdot s}\right] \tag{5.7}$$

The rate of emission from a 1 m^2 column along the line of sight is normally just $4\pi L_\gamma$ for any isotropic source with no self-absorption.

Then the column emission rate in Rayleigh can be obtained from the L_γ :

$$R = \frac{4\pi L_\gamma}{10^{10}}[Rayleighs] \tag{5.8}$$

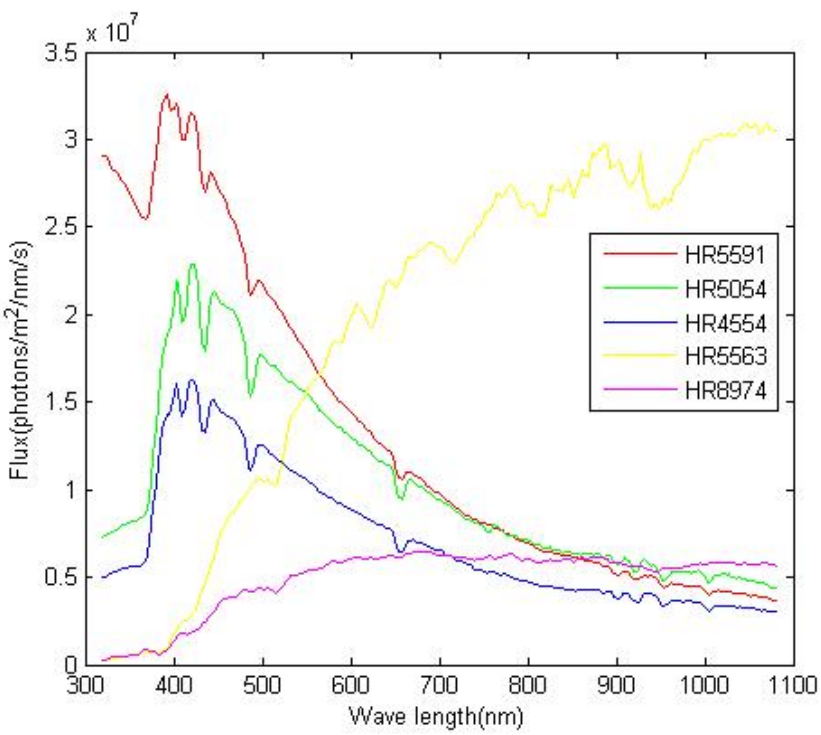


Figure 5-5 Flux for five stars in quantum units

5.3.2. Atmospheric absorption

The emission lines observed by ALIS at 4278Å, 5577Å, 6300Å are absorbed mainly in the troposphere and stratosphere; they are all optically thin in the region of auroral emission, i.e. no incident radiant energy is absorbed by the ionosphere --. Mainly absorption is caused by stratospheric ozone. The amount of absorption is up to the length of the light path through the stratosphere. Thus the smallest amount of absorption is in the direction with zero zenith angles, and the absorption increases along with increasing the zenith angle (*Gustavsson*, 2000).

The optical depth expresses the quantity of how much light absorbed or scattered in passing through a medium. If I_0 is the intensity of incoming light and I is the observed intensity after a given path, then optical depth τ is defined by the following equation (*Kitchin, Christopher Robert,* 1987):

$$\frac{I}{I_0} = e^{-\tau} \tag{5.9}$$

Taking the zenith angle and wavelength into account, the absorption is (*Gustavsson*, 2000):

$$\frac{I}{I_0} = e^{-\tau(\lambda)/\cos\theta_z} \tag{5.10}$$

Table 5-3 shows the optical depths by *Arnoldy and Lewis* (1976) that are used for aroral line measurements by ALIS cameras. And variations of optical depth result from variance of zenith angle for 5 stars are summarized in Table 5-4.

λ (Å)	τ
4278	0.236
5577	0.116
6300	0.081

Table 5-3 Optical depths for auroral lines (*Gustavsson*, 2000)

HIP	Star name	Zenith angle(degree)	Amount of absorption		
			red	green	blue
4554	Gamma UMa	17-15	0.9189-0.9196	0.8859-0.8869	0.7815-0.7834
5054	Zeta UMa	22.5-18	0.9160-0.9184	0.8819-0.8853	0.7744-0.7804
5191	Eta UMa	29-24.5	0.9113-0.9148	0.8755-0.8803	0.7629-0.7715
5563	Beta UMi	11-21	0.9169-0.9208	0.8832-0.8885	0.7766-0.7863
8974	Gamma Cep	34-35	0.9058-0.9069	0.8680-0.8694	0.7497-0.7523

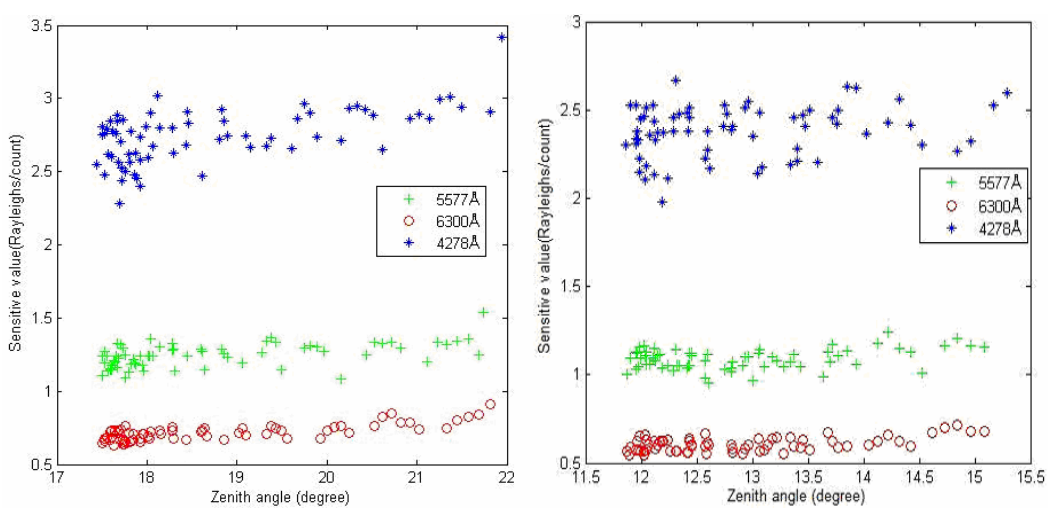
Table 5-4 Variance of zenith angle for five stars in one hour on date 2011-3-14.The variance of zenith angle is less than 5 degree in one hour and leads to small change in optical depths. So it is considered that the amount of absorption for the same star taken by the same camera is constant.

Thus, we can calculate the sensitive value $C_{abs\lambda c}$ by:

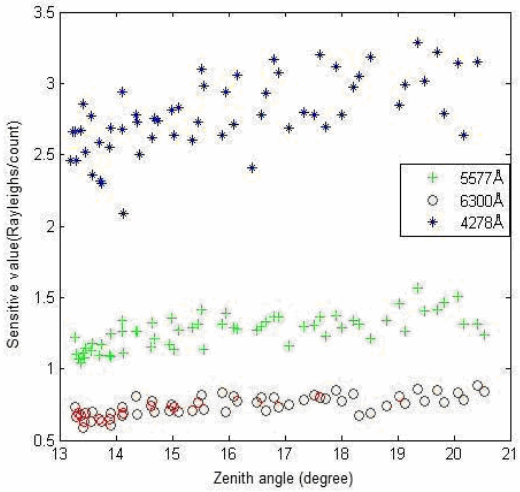
$$C_{abs\lambda c} = \frac{I_{cal} \cdot e^{-\tau(\lambda)/\cos\theta_z} \cdot t_{\text{int}}}{DN} \left[\frac{Rayleighs}{Count} \right]$$

Where, I_{cal} is obtained by the integration of the stellar spectrum over the filter pass bandwidth, the calculation results for five stars against their zenith angle have been plotted in Figure 5-6.

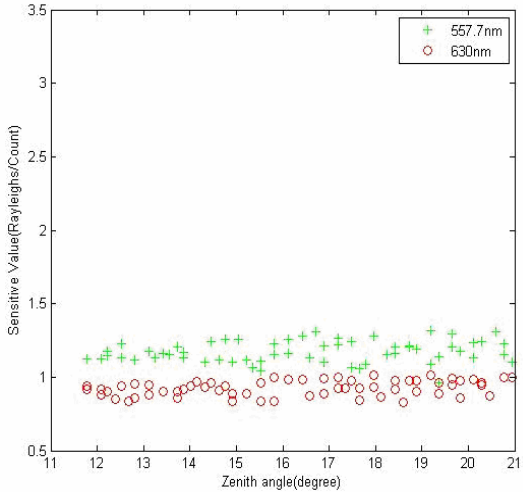
DN is the digital counts produced by luminary.



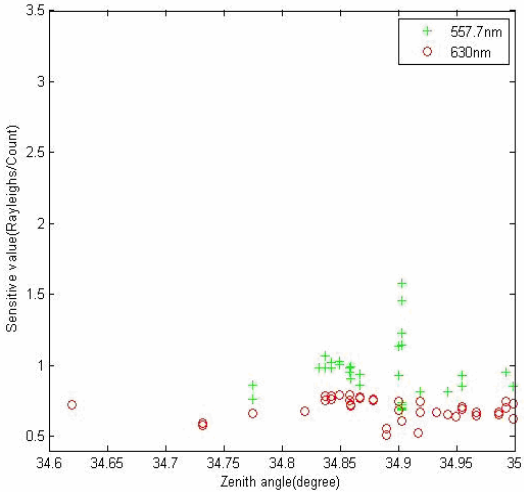
A. HR 5191 (Camera 5) B. HR 5054 (Camera 5)



C. HR 4554 (Camera 5)



D.HR 5563 (Camera 3)



E. HR 8974 (Camera 3)

Figure 5-6 $C_{abs\lambda c}$ against zenith angle for five stars

Results from the stellar absolute calibration are listed in the Table 5-5 and Table 5-6.

V magnitude	HIP	Star name	4278 Å [R/Count]	5577 Å [R/Count]	6300 Å [R/Count]
2.44	4554	Gamma UMa	2.7879	1.2369	0.7406
2.06	5054	Zeta UMa	2.3802	1.0876	0.6056
1.86	5191	Eta UMa	2.7448	1.2457	0.7161

Table 5-5 Absolute calibration results by using stellar spectrum, for camera 5

V magnitude	HIP	Star name	4278 Å [R/Count]	5577 Å [R/Count]	6300 Å [R/Count]
2.08	5563	Beta UMi	null	1.1727	0.9218
3.21	8974	Gamma Cep	null	0.9259	0.6921

Table 5-6 Absolute calibration results by using stellar spectrum, for camera 3. No results from line 4278 Å due to low SNR.

5.4. Summary

As can be seen from Table 5-5 and Table 5-6, for same camera, the values of C_{abs} of different stars, but with the same wavelength, are approximately equivalent. Comparing to laboratory results (see Table 5-7 and Table 5-8), it is not difficult to find that for camera 5, 0.72 times difference exists for red line, and for green and blue lines the difference is approximately 0.95 and 1.05 times, respectively; for camera 3, only results form lines 6300 Å and 5577 Å due to low SNR, and the ratio are 0.97 and 1.29 respectively. There are several reasons cause these differences: 1. The error between the calculated absorption by using theoretical optical depth and the real atmospheric attenuation. 2. The imprecise of calculating star counts caused by noise. 3. For statistic aspect, there are only samples from 3 hours in one day, and only images from bin 4×4 case, for a further investigation, more samples are needed in different operational modes.

λ (Å)	Stellar Calibration	Laboratory Calibration	Ratio
	[Rayleighs/(Count/s)]		
4278	2.6376	2.5135	1.05
5577	1.1901	1.2580	0.95
6300	0.6874	0.9522	0.72

Table 5-7 A comparison between the results of stellar calibration and laboratory calibration, results from stellar calibration in column two and three from laboratory calibrations (Camera 5)

λ (Å)	Stellar Calibration	Laboratory Calibration	Ratio
	[Rayleighs/(Count/s)]		
5577	1.0493	0.8111	1.29
6300	0.8090	0.8273	0.97

Table 5-8 A comparison between the results of stellar calibration and laboratory calibration, results from stellar calibration in column two and three from laboratory calibrations (Camera 3)

Because of limited numbers of data, this stellar absolute calibration has been done by using only 5 stars. For each star and each filter, only 90 samples are used for calculating the integral of digital counts. More calibrator samples are required to obtain a greater statistical result.

Chapter 6

Conclusion

This thesis focus on studying ALIS imagers absolute calibration using laboratory light-standards and well known stellar spectrums. Because the techniques using stellar spectrum take the atmosphere and domes into account automatically, the difference of results from two light sources has been found and discussed.

The variations caused by pixel-to-pixel sensitivity are removed by unifying the product of field-of-view and effective area of each pixel. Field-of-view of each pixel is determined by the optical characteristics of the camera. An improved camera model (*Gustavsson*, 2000) is used for determining optical parameters. Compared to the pin hole camera model, it appears that radial error has been removed by using this modified model. Optical parameters found from optimization fit between stars positions from the YBS star catalogue and the star positions from in night sky images.

Former calibration, conducted in 1997 (*Brändström*, 2003), assumed 6 ALIS imagers with complete linearity. However, after a linearity test for each camera by using laboratory data in Sept. 2007, it was found that cameras exhibit non-linearity as pixel intensity over a certain value, especially for camera 2. Thus, non-linearity correction is introduced in the camera signature removing procedure. Analysis of experimental results shows that linearity has been improved by using a quadratic function. The light standards are inter-calibrated against other light-standards during European calibration workshops, and then pixel sensitivities are calculated for different imagers with different gain, binning, exposure time and light source. There are five sets of sensitivity values for five ALIS imagers included in appendix (Camera 2-Camera 6, Camera 1 is an exception in this calibration because of hardware problem for one quadrant), it is clearly shown that calibration results appear very stable with different gain, exposure time and light source.

And for different binning, the sensitivity value is in a four times relationship, so, we conclude that our calibration result by using laboratory data is quite reasonable.

To know the effect of natural environment, for example, atmosphere and domes, absolute stellar calibration is carried out for camera 3 and camera 5. Light standards are chosen from well-known stellar spectrum, thus the effect of atmospheric absorption and domes are automatically taken into account. Optimization tools are chosen for fitting background and stellar PSF functions. After eliminating the background, digital numbers produced by star are calculated by integrating the PSF function. Comparing the sensitivity values calculated by using stellar spectrum and laboratory light standard, it is found that there are 0.72 and 0.97 times differences for red line (630nm), and the difference ratio increases as the light wavelength decreases. For a further investigation, it is needed that more samples from different time and different camera operational modes.

Appendix A

Calibration results for Camera2-Camera6 in 2011, by using laboratory data from Sept.2007. The coefficient C are given in [R/count], for each gain, bin and light standards. Column one describes the exposure times (unit in seconds), numbers in bracket are times for lamp Lauch. The last column shows the pixels are selected for calibration, in a digital output matrix.Lamps' name, Y275: IRF, UJO Y275, Lauch: IRF, Lauche lamp, 920B: IRF, UJO 920B.

Camera 2

Camera2, 5577 Å, Bin:1×1							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:402:482(Y275)
10(50)	23.2	23.7	23.0	23.5	22.9	23.3	Y:516:596
20(100)	22.9	23.4	22.8	23.3	22.9	23.3	
30(150)	22.6	23.1	22.5	23.0	22.9	23.3	X:426:506(Lauch)
40(200)	22.8	23.3	22.6	23.1	22.8	23.3	Y:528:608
Average	22.9	23.4	22.7	23.2	22.9	23.3	23.1

Camera2, 5577 Å, Bin:2×2							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:191:251(Y275)
2(10)	5.7	5.9	5.7	5.9	5.6	5.8	Y:248:308
5(25)	5.7	5.8	5.6	5.8	5.7	5.8	
10(50)	5.7	5.8	5.7	5.8	5.7	5.8	X:203:263(Lauch)
20(100)	5.7	5.8	5.7	5.9	5.7	5.8	Y:254:314
Average	5.7	5.8	5.7	5.9	5.7	5.8	5.8

Camera2, 5577 Å, Bin:4×4							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:96:126(Y275)
2(5)	1.34	1.42	1.34	1.42	1.37	1.43	Y:124:154
3(15)	1.37	1.41	1.37	1.41	1.37	1.42	
4(20)	1.36	1.41	1.37	1.41	1.36	1.41	X:101:131(Lauch)
5(25)	1.33	1.39	1.34	1.39	1.35	1.39	Y:127:157
Average	1.35	1.41	1.36	1.41	1.36	1.41	1.38

Camera2, 6300 Å, Bin:1×1							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:402:482(Y275)
10(10)	18.2	18.9	18.1	18.7	18.1	18.6	Y:516:596
20(20)	17.9	18.6	17.8	18.5	18.0	18.6	
30(30)	17.9	18.4	17.8	18.3	18.0	18.6	X:426:506(Lauch)
40(40)	18.2	18.6	18.1	18.5	18.0	18.5	Y:528:608
Average	18.0	18.6	18.0	18.5	18.0	18.6	18.3

Camera2, 6300 Å, Bin:2×2							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:191:251(Y275)
2(2)	4.5	4.6	4.4	4.6	4.4	4.6	Y:248:308
5(5)	4.4	4.6	4.4	4.6	4.5	4.6	
10(10)	4.5	4.6	4.5	4.6	4.5	4.6	X:203:263(Lauch)
20(20)	4.3	4.6	4.3	4.6	4.4	4.6	Y:254:314
Average	4.4	4.6	4.4	4.6	4.5	4.6	4.5

Camera2, 6300 Å, Bin:4×4							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:96:126(Y275)
2(2)	1.06	1.09	1.06	1.09	1.07	1.11	Y:124:154
3(3)	1.08	1.11	1.08	1.11	1.07	1.11	
4(4)	1.05	1.10	1.05	1.11	1.06	1.11	X:101:131(Lauch)
5(5)	1.00	1.07	1.01	1.08	1.04	1.10	Y:127:157
Average	1.05	1.09	1.05	1.10	1.06	1.11	1.08

Camera2, 4278 Å, Bin: 1×1				Pixels selected
Gain	13	20	31	
Lamp	920B	920B	920B	X:438:518
30s	41.7	41.3	41.2	Y:500:580
60s	41.3	40.9	41.2	
120s	40.8	40.4	41.2	
200s	41.8	41.4	41.2	
Average	41.4	41.0	41.2	41.2

Camera2, 4278 Å, Bin: 2×2				Pixels selected
gain	13	20	31	
Lamp	920B	920B	920B	X:209:269
10s	10.3	10.3	10.3	Y:240:300
20s	10.2	10.2	10.3	
30s	10.2	10.1	10.3	
60s	10.4	10.4	10.3	
Average	10.3	10.3	10.3	10.3

Camera2, 4278 Å, Bin: 4×4				Pixels selected
gain	13	20	31	
lamp	920B	920B	920B	X:105:135
2.5s	2.49	2.48	2.50	Y:120:150
5s	2.47	2.46	2.51	
10s	2.49	2.48	2.50	
15s	2.46	2.46	2.48	
Average	2.48	2.47	2.50	2.48

Camera 3

Camera3, 5577 Å, Bin:1×1							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:552:632(Y275)
10(50)	13.4	13.9	13.2	13.6	13.0	13.3	Y:511:591
20(100)	13.3	13.8	13.1	13.6	12.9	13.3	
30(150)	13.2	13.6	13.1	13.6	12.9	13.3	X:510:590(Lauch)
40(200)	13.5	13.9	13.2	13.6	13.0	13.4	Y:523:603
Average	13.4	13.8	13.2	13.6	13.0	13.3	13.4

Camera3, 5577 Å, Bin:2×2							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:266:326(Y275)
2(10)	3.2	3.4	3.2	3.4	3.2	3.4	Y:246:306
5(25)	3.2	3.3	3.3	3.4	3.2	3.4	
10(50)	3.3	3.4	3.3	3.4	3.3	3.4	X:245:305(Lauch)
20(100)	3.2	3.3	3.2	3.4	3.2	3.3	Y:252:312
Average	3.2	3.4	3.3	3.4	3.2	3.4	3.3

Camera3, 5577 Å, Bin:4×4							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:133:163(Y275)
2(5)	0.78	0.83	0.81	0.84	0.79	0.83	Y:123:153
3(15)	0.81	0.84	0.82	0.85	0.79	0.83	
4(20)	0.79	0.83	0.81	0.85	0.79	0.82	X:123:153(Lauch)
5(25)	0.75	0.79	0.80	0.84	0.77	0.80	Y:126:156
Average	0.78	0.82	0.81	0.84	0.79	0.82	0.81

Camera3, 6300 Å, Bin:1×1							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:552:632(Y275)
10(10)	13.8	14.3	13.5	14.0	13.3	13.7	Y:511:591
20(20)	13.7	14.2	13.5	14.0	13.3	13.7	
30(30)	13.6	14.1	13.5	13.9	13.3	13.7	X:510:590(Lauch)
40(40)	13.8	14.0	13.5	13.9	13.3	13.7	Y:523:603
Average	13.7	14.2	13.5	14.0	13.3	13.7	13.7

Camera3, 6300 Å, Bin:2×2							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:266:326(Y275)
2(2)	3.4	3.5	3.3	3.4	3.3	3.4	Y:246:306
5(5)	3.4	3.5	3.4	3.5	3.3	3.4	
10(10)	3.4	3.5	3.4	3.5	3.3	3.4	X:245:305(Lauch)
20(20)	3.3	3.5	3.3	3.5	3.3	3.4	Y:252:312
Average	3.4	3.5	3.4	3.5	3.3	3.4	3.4

Camera3, 6300 Å, Bin:4×4							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:133:163(Y275)
2(2)	0.81	0.83	0.82	0.85	0.81	0.84	Y:123:153
3(3)	0.83	0.85	0.82	0.85	0.82	0.84	
4(4)	0.82	0.86	0.82	0.85	0.81	0.84	X:123:153(Lauch)
5(5)	0.78	0.84	0.80	0.84	0.80	0.83	Y:126:156
Average	0.81	0.85	0.82	0.85	0.81	0.84	0.83

Camera3, 4278 Å, Bin: 1×1				Pixels selected
Gain	10	21	32	
Lamp	920B	920B	920B	X:504:584
30s	28.8	28.2	27.7	Y:510:590
60s	28.7	28.2	27.7	
120s	28.2	28.1	27.6	
200s	29.3	28.3	27.8	
Average	28.8	28.2	27.7	28.2

Camera3, 4278 Å, Bin: 2×2				Pixels selected
Gain	13	20	31	
lamp	920B	920B	920B	X:242:302
10s	7.1	7.1	7.0	Y:245:305
20s	7.1	7.1	7.0	
30s	7.0	7.0	7.0	
60s	7.3	7.1	7.0	
Average	7.1	7.1	7.0	7.1

Camera3, 4278 Å, Bin: 4×4				Pixels selected
gain	13	20	31	
lamp	920B	920B	920B	X:121:151
2.5s	1.74	1.73	1.72	Y:123:153
5s	1.73	1.74	1.72	
10s	1.73	1.74	1.72	
15s	1.74	1.72	1.71	
Average	1.74	1.73	1.72	1.73

Camera 4

Camera4, 5577 Å, Bin:1×1							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:539:639(Y275)
10(50)	16.0	16.4	15.4	15.8	15.2	15.5	Y:405:505
20(100)	16.0	16.3	15.4	15.8	15.2	15.5	
30(150)	16.0	16.3	15.4	15.8	15.1	15.5	X:509:609(Lauch)
40(200)	16.0	16.3	15.4	15.8	15.2	15.5	Y:419:519
Average	16.0	16.3	15.4	15.8	15.2	15.5	15.7

Camera4, 5577 Å, Bin:2×2							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:264:324(Y275)
2(10)	3.9	4.0	3.8	3.9	3.8	3.9	Y:198:258
5(25)	3.9	4.0	3.8	3.9	3.8	3.9	
10(50)	3.9	4.0	3.8	3.9	3.8	3.9	X:249:309(Lauch)
20(100)	3.9	4.0	3.8	3.9	3.8	3.9	Y:205:265
Average	3.9	4.0	3.8	3.9	3.8	3.9	3.9

Camera4, 5577 Å, Bin:4×4							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:132:162(Y275)
2(5)	0.93	0.98	0.93	0.96	0.92	0.95	Y:99:129
3(15)	0.95	0.98	0.93	0.96	0.92	0.95	
4(20)	0.95	0.98	0.93	0.96	0.92	0.95	X:125:155(Lauch)
5(25)	0.92	0.96	0.91	0.94	0.90	0.93	Y:103:133
Average	0.94	0.98	0.93	0.96	0.92	0.95	0.95

Camera4, 6300 Å, Bin:1×1							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:539:639(Y275)
10(10)	16.4	16.6	15.8	16.0	15.5	15.7	Y:405:505
20(20)	16.3	16.6	15.8	16.0	15.5	15.7	
30(30)	16.2	16.5	15.8	16.0	15.5	15.7	X:509:609(Lauch)
40(40)	16.3	16.4	15.8	16.0	15.5	15.7	Y:419:519
Average	16.3	16.5	15.8	16.0	15.5	15.7	16.0

Camera4, 6300 Å, Bin:2×2							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:264:324(Y275)
2(2)	4.0	4.1	3.9	3.9	3.8	3.9	Y:198:258
5(5)	4.0	4.1	3.9	4.0	3.9	3.9	
10(10)	4.0	4.0	3.9	4.0	3.9	3.9	X:249:309(Lauch)
20(20)	4.0	4.1	3.9	4.0	3.9	3.9	Y:205:265
Average	4.0	4.1	3.9	4.0	3.9	3.9	4.0

Camera4, 6300 Å, Bin:4×4							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:132:162(Y275) Y:99:129 X:125:155(Lauch) Y:103:133
2(2)	0.96	0.97	0.95	0.96	0.94	0.95	
3(3)	0.97	0.98	0.95	0.97	0.94	0.95	
4(4)	0.98	0.99	0.95	0.97	0.94	0.95	
5(5)	0.96	0.98	0.94	0.96	0.93	0.95	
Average	0.97	0.98	0.95	0.97	0.94	0.95	0.96

Camera4, 4278 Å, Bin: 1×1				Pixels selected
Gain	10	21	32	
Lamp	920B	920B	920B	X:498:578 Y:441:521
30s	33.7	32.5	31.8	
60s	33.6	32.5	31.8	
120s	33.3	32.4	31.8	
200s	34.3	32.6	31.9	
Average	33.7	32.5	31.8	32.7

Camera4, 4278 Å, Bin: 2×2				Pixels selected
gain	10	21	32	
lamp	920B	920B	920B	X:239:299 Y:211:271
10s	8.3	8.1	8.0	
20s	8.3	8.1	8.0	
30s	8.2	8.1	8.0	
60s	8.5	8.1	8.0	
Average	8.3	8.1	8.0	8.1

Camera4, 4278 Å, Bin: 4×4				Pixels selected
gain	10	20	32	
lamp	920B	920B	920B	X:120:150 Y:105:135
2.5s	2.01	1.97	1.94	
5s	2.01	1.99	1.96	
10s	1.99	1.98	1.96	
15s	2.03	1.98	1.95	
Average	2.01	1.98	1.95	1.98

Camera 5

Camera5, 5577 Å, Bin:1×1							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:509:609(Y275) Y:453:553 X:494:594(Lauch) Y:485:585
10(50)	20.1	21.1	19.7	20.6	19.7	20.4	
20(100)	20.2	21.1	19.7	20.6	19.7	20.5	
30(150)	20.2	21.1	19.7	20.6	19.6	20.5	
40(200)	20.3	21.2	19.8	20.7	19.7	20.5	
Average	20.2	21.1	19.7	20.6	19.7	20.5	20.3

Camera5, 5577 Å, Bin:2×2							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:245:315(Y275) Y:217:287 X:233:303(Lauch) Y:223:293
2(10)s	5.0	5.3	4.9	5.2	4.9	5.1	
5(25)s	5.0	5.3	4.9	5.2	4.9	5.1	
10(50)s	5.0	5.3	4.9	5.2	4.9	5.2	
20(100)s	5.2	5.4	5.0	5.2	5.0	5.2	
Average	5.1	5.3	4.9	5.2	4.9	5.2	5.1

Camera5, 5577 Å, Bin:4×4							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:120:160(Y275) Y:106:146 X:114:154(Lauch) Y:109:149
2(5)s	1.24	1.31	1.21	1.28	1.21	1.28	
3(15)s	1.24	1.30	1.21	1.28	1.21	1.28	
4(20)s	1.25	1.31	1.21	1.28	1.21	1.27	
5(25)s	1.27	1.33	1.23	1.29	1.21	1.27	
Average	1.25	1.31	1.22	1.28	1.21	1.28	1.26

Camera5, 6300 Å, Bin:1×1							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:509:609(Y275) Y:453:553 X:494:594(Lauch) Y:485:585
10(10)s	15.4	15.9	15.0	15.5	15.0	15.4	
20(20)s	15.4	15.9	15.0	15.5	15.0	15.4	
30(30)s	15.5	15.9	15.1	15.5	15.0	15.4	
40(40)s	15.5	15.9	15.1	15.6	15.0	15.5	
Average	15.5	15.9	15.1	15.5	15.0	15.4	15.4

Camera5, 6300 Å, Bin:2×2							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:245:315(Y275) Y:217:287 X:233:303(Lauch) Y:223:293
2(2)s	3.8	3.9	3.7	3.9	3.7	3.8	
5(5)s	3.8	4.0	3.8	3.9	3.8	3.9	
10(10)s	3.8	4.0	3.8	3.9	3.8	3.9	
20(20)s	3.9	4.1	3.9	4.0	3.8	3.9	
Average	3.8	4.0	3.8	3.9	3.8	3.9	3.9

Camera5, 6300 Å, Bin:4×4							Pixels selected
gain	13		20		31		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:120:160(Y275) Y:106:146 X:114:154(Lauch) Y:109:149
2(2)s	0.94	0.98	0.93	0.96	0.93	0.95	
3(3)s	0.95	0.98	0.92	0.96	0.92	0.96	
4(4)s	0.97	0.99	0.94	0.96	0.92	0.95	
5(5)s	0.96	1.00	0.95	0.98	0.92	0.95	
Average	0.96	0.99	0.94	0.96	0.92	0.95	0.95

Camera5, 4278 Å, Bin: 1×1				Pixels selected
Gain	13	20	31	
Lamp	920B	920B	920B	X:497:597 Y:442:542
30s	41.1	40.1	39.7	
60s	41.1	40.1	39.7	
120s	41.2	40.1	39.8	
200s	41.4	40.3	39.9	
Average	41.2	40.2	39.8	40.4

Camera5, 4278 Å, Bin: 2×2				Pixels selected
gain	13	20	31	
lamp	920B	920B	920B	X:238:308 Y:211:281
10s	10.3	10.1	10.1	
20s	10.3	10.1	10.1	
30s	10.3	10.1	10.1	
60s	10.4	10.1	10.1	
Average	10.3	10.1	10.1	10.2

Camera5, 4278 Å, Bin: 4×4				Pixels selected
gain	13	20	31	
lamp	920B	920B	920B	X:116:156 Y:103:143
2.5s	2.55	2.49	2.49	
5s	2.55	2.50	2.50	
10s	2.55	2.50	2.50	
15s	2.56	2.49	2.49	
Average	2.55	2.50	2.50	2.52

Camera 6

Camera6, 5577 Å, Bin:1×1							Pixels selected
Gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:560:660(Y275)
10(50)	14.2	14.7	14.0	14.4	13.9	14.3	Y:412:512
20(100)	14.2	14.7	14.0	14.4	13.9	14.3	
30(150)	14.2	14.7	14.0	14.4	13.9	14.3	X:458:558(Lauch)
40(200)	14.2	14.7	14.0	14.4	13.8	14.2	Y:433:533
Average	14.2	14.7	14.0	14.4	13.9	14.3	14.3

Camera6, 5577 Å, Bin:2×2							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:275:335(Y275)
2(10)	3.5	3.7	3.4	3.6	3.4	3.6	Y:201:261
5(25)	3.5	3.7	3.5	3.6	3.4	3.6	
10(50)	3.5	3.7	3.5	3.6	3.5	3.6	X:224:284(Lauch)
20(100)	3.5	3.7	3.5	3.6	3.5	3.6	Y:212:272
Average	3.5	3.7	3.5	3.6	3.5	3.6	3.6

Camera6, 5577 Å, Bin:4×4							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:133:173(Y275)
2(5)	0.87	0.92	0.86	0.90	0.85	0.90	Y:96:136
3(15)	0.87	0.91	0.86	0.90	0.85	0.90	
4(20)	0.87	0.91	0.86	0.90	0.86	0.90	X:107:14(Lauch)
5(25)	0.87	0.92	0.86	0.90	0.86	0.89	Y:101:141
Average	0.87	0.92	0.86	0.90	0.86	0.90	0.89

Camera6, 6300 Å, Bin:1×1							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:560:660(Y275)
10(10)	10.4	11.0	10.3	10.8	10.2	10.7	Y:412:512
20(20)	10.4	11.0	10.3	10.8	10.2	10.7	
30(30)	10.4	11.0	10.3	10.8	10.2	10.7	X:458:558(Lauch)
40(40)	10.4	11.0	10.3	10.8	10.1	10.6	Y:433:533
Average	10.4	11.0	10.3	10.8	10.2	10.7	10.6

Camera6, 6300 Å, Bin:2×2							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:275:335(Y275)
2(2)	2.5	2.7	2.5	2.6	2.5	2.6	Y:201:261
5(5)	2.6	2.7	2.5	2.7	2.5	2.7	
10(10)	2.6	2.7	2.5	2.7	2.5	2.7	X:224:284(Lauch)
20(20)	2.7	2.8	2.7	2.7	2.7	2.7	Y:212:272
Average	2.6	2.7	2.6	2.7	2.6	2.7	2.7

Camera6, 6300 Å, Bin:4×4							Pixels selected
gain	10		21		32		
Lamp(s)	Y275	Lauch	Y275	Lauch	Y275	Lauch	X:133:173(Y275) Y:96:136 X:107:14(Lauch) Y:101:141
2(2)	0.64	0.67	0.63	0.66	0.62	0.66	
3(3)	0.64	0.68	0.63	0.67	0.63	0.66	
4(4)	0.64	0.68	0.63	0.67	0.63	0.66	
5(5)	0.67	0.68	0.66	0.67	0.66	0.67	
Average	0.65	0.68	0.64	0.67	0.64	0.66	0.66

Camera6, 4278 Å, Bin: 1×1				Pixels selected
Gain	10	21	32	
Lamp	920B	920B	920B	X:443:543 Y:440:540
30s	22.9	22.5	22.3	
60s	22.9	22.5	22.3	
120s	22.9	22.5	22.2	
200s	23.0	22.5	22.1	
Average	22.9	22.5	22.2	22.5

Camera6, 4278 Å, Bin: 2×2				Pixels selected
gain	10	21	32	
lamp	920B	920B	920B	X:227:267 Y:225:265
10s	5.7	5.6	5.6	
20s	5.7	5.6	5.6	
30s	5.7	5.6	5.6	
60s	5.7	5.6	5.6	
Average	5.7	5.6	5.6	5.6

Camera6, 4278 Å, Bin: 4×4				Pixels selected
gain	10	21	32	
lamp	920B	920B	920B	X:110:140 Y:106:136
2.5s	1.42	1.39	1.39	
5s	1.43	1.40	1.40	
10s	1.42	1.40	1.39	
15s	1.43	1.40	1.39	
Average	1.43	1.40	1.39	1.41

Appendix B

Coefficient for correction fuction b, c (Chapter 4, Section 4.2.3).

$$DN'_{(\mathrm{int})i,j} = DN_{(\mathrm{int})i,j} \cdot \left(1 + \left(b \cdot DN_{(\mathrm{int})i,j} + c\right)\right)$$

Camera2, Bin:1×1						
gain	13		21		31	
Quadrant	B	C	b	C	B	C
0	3.942e-6	1.979e-4	3.501e-6	1.532e-4	6.060e-7	1.425e-4
1	2.621e-6	3.241e-4	2.331e-6	1.987e-4	4.660e-7	1.863e-4
2	3.221e-6	8.533e-5	3.075e-6	1.125e-4	5.755e-7	1.640e-4
3	2.781e-6	2.006e-4	2.395e-6	1.207e-4	5.661e-7	8.228e-4

Camera2, Bin:2×2						
gain	13		21		31	
Quadrant	B	C	b	C	B	C
0	3.958e-6	1.809e-4	3.577e-6	1.633e-4	5.851e-7	1.809e-4
1	2.298e-6	1.052e-4	2.252e-6	2.477e-4	4.890e-7	1.052e-4
2	3.136e-6	1.652e-4	2.866e-6	1.707e-4	6.189e-7	1.652e-4
3	2.582e-6	1.570e-4	2.334e-6	2.090e-4	5.988e-7	1.507e-4

Camera2, Bin:4×4						
gain	13		21		31	
Quadrant	b	C	b	C	B	C
0	5.600e-6	1.532e-4	5.286e-6	1.433e-4	1.753e-6	1.532e-4
1	4.130e-6	2.477e-4	4.291e-6	1.986e-4	1.691e-6	2.477e-4
2	4.706e-6	1.707e-4	4.510e-6	1.824e-4	1.854e-6	1.707e-4
3	4.630e-6	1.105e-4	4.339e-6	1.853e-4	1.924e-6	1.105e-4

Camera3, Bin:1×1						
gain	10		21		32	
Quadrant	b	C	b	c	B	C
0	2.238e-6	2.648e-4	4.026e-7	8.361e-5	4.821e-7	1.903e-4
1	1.537e-6	1.019e-4	3.445e-7	2.185e-4	3.826e-7	1.355e-4
2	1.398e-6	9.639e-5	3.482e-7	2.093e-4	3.997e-7	1.422e-4
3	1.367e-6	8.999e-5	2.903e-7	1.290e-5	3.481e-7	2.068e-4

Camera3,Bin:2×2						
gain	10		21		32	
Quadrant	b	C	B	c	B	C
0	2.198e-6	8.979e-5	3.780e-7	1.981e-4	3.744e-7	1.261e-4
1	1.538e-6	3.847e-5	2.710e-7	2.271e-4	2.764e-7	2.568e-4
2	1.374e-6	1.607e-4	2.770e-7	1.330e-4	2.888e-7	1.652e-4
3	1.420e-6	2.016e-4	2.698e-7	5.848e-5	2.702e-7	1.413e-4

Camera3,Bin:4×4						
gain	10		21		32	
Quadrant	B	C	B	c	B	C
0	3.399e-6	2.312e-4	9.914e-7	1.369e-4	9.613e-7	2.665e-4
1	2.537e-6	1.292e-4	8.019e-7	7.178e-5	8.258e-7	2.027e-4
2	2.335e-6	1.337e-4	8.672e-7	1.368e-4	8.319e-7	1.954e-4
3	2.433e-6	1.548e-4	7.970e-7	2.330e-4	8.003e-7	9.792e-5

Camera4,Bin:1×1						
gain	10		21		32	
Quadrant	b	C	b	c	b	C
0	1.844e-6	1.924e-4	3.906e-7	7.976e-5	4.033e-7	1.026e-4
1	8.479e-7	1.265e-4	3.029e-7	8.905e-5	3.495e-7	2.072e-4
2	8.867e-7	2.310e-4	3.097e-7	1.019e-4	3.282e-7	1.761e-4
3	1.181e-6	1.677e-4	3.237e-7	1.665e-4	3.654e-7	2.725e-5

Camera4,Bin:2×2						
gain	10		21		32	
Quadrant	b	C	b	c	b	C
0	2.084e-6	1.508e-4	4.255e-7	1.333e-4	3.913e-7	1.625e-4
1	1.372e-6	1.533e-4	3.574e-7	1.658e-4	3.317e-7	2.645e-4
2	1.228e-6	4.386e-5	3.304e-7	1. 258e-4	3.076e-7	2.419e-4
3	1.481e-6	1.354e-4	3.497e-7	2.004e-4	3.208e-7	1.907e-4

Camera4,Bin:4×4						
gain	10		21		32	
Quadrant	b	C	b	c	b	C
0	3.410e-6	1.184e-4	1.090e-6	8.361e-5	1.064e-6	1.939e-4
1	2.399e-6	9.174e-5	9.986e-7	2.185e-4	9.920e-7	2.129e-4
2	2.180e-6	8.878e-5	9.186e-7	2.093e-4	8.960e-7	6.748e-5
3	2.614e-6	1.283e-4	9.853e-7	1.290e-4	9.826e-7	1.833e-4

Camera5,Bin:1×1						
gain	13		20		31	
Quadrant	b	c	b	c	b	C
0	3.815e-8	1.940e-4	-1.741e-7	1.189e-4	-5.140e-9	2.657e-4
1	-4.181e-7	2.111e-4	-2.096e-7	2.353e-4	-1.7886e-8	1.677e-4
2	-1.725e-7	2.656e-4	-1.545e-7	1.432e-4	2.175e-8	2.225e-4
3	-1.744e-7	6.821e-5	-1.1559e-7	1.414e-4	8.454e-8	1.086e-4

Camera5,Bin:2×2						
gain	13		20		31	
Quadrant	b	c	b	c	b	C
0	8.293e-7	8.591e-5	-2.064e-8	2.123e-4	-1.009e-7	1.188e-4
1	3.556e-9	3.274e-4	-9.661e-8	1.667e-4	-8.006e-8	1.962e-4
2	4.634e-8	1.455e-4	-1.491e-7	1.019e-4	-1.204e-7	5.913e-5
3	7.885e-8	1.607e-4	-4.838e-8	1.204e-4	-3.926e-8	1.142e-4

Camera5,Bin:4×4						
gain	13		20		31	
Quadrant	b	c	b	c	b	C
0	1.371e-6	1.184e-4	5.401e-7	1.793e-4	4.654e-7	2.049e-4
1	3.452e-7	2.412e-4	4.329e-7	2.392e-4	3.874e-7	2.030e-4
2	4.746e-7	2.049e-4	3.619e-7	1.819e-4	3.577e-7	2.058e-4
3	4.385e-7	1.695e-4	5.182e-7	2.063e-4	3.922e-7	2.028e-4

Camera6,Bin:1×1						
gain	10		21		32	
Quadrant	b	C	b	C	B	C
0	2.612e-7	1.273e-4	7.609e-8	1.111e-4	4.033e-7	1.026e-4
1	7.461e-8	2.601e-5	1.027e-7	1.657e-5	3.495e-7	2.072e-4
2	8.428e-8	1.033e-4	6.843e-8	2.264e-4	3.282e-7	1.761e-4
3	1.059e-7	1.003e-4	1.160e-7	1.111e-4	3.654e-7	2.725e-5

Camera6,Bin:2×2						
gain	10		21		32	
Quadrant	b	C	b	C	B	C
0	3.819e-7	1.735e-4	1.710e-6	2.279e-4	1.757e-7	2.460e-4
1	2.170e-7	1.524e-4	2.061e-6	2.140e-4	1.892e-7	1.410e-4
2	1.664e-7	1.562e-4	1.669e-6	1.687e-4	1.582e-7	1.547e-4
3	2.107e-7	2.136e-4	2.039e-6	1.661e-4	1.998e-7	2.218e-4

Camera6,Bin:4×4						
gain	10		21		32	
Quadrant	b	C	b	C	B	C
0	7.228e-7	1.854e-4	3.204e-7	1.723e-4	3.260e-7	1.333e-4
1	3.912e-7	1.179e-4	3.336e-7	2.102e-4	3.363e-7	2.950e-4
2	3.045e-7	1.392e-4	2.773e-7	1.486e-4	2.789e-7	1.578e-4
3	3.602e-7	1.193e-4	3.248e-7	2.645e-4	3.270e-7	1.724e-4

Bibliography

Alekseeca G.A and A.A. Arkharov. The Pulkovo Spectrophotometric Catalog of Bright Stars in the Range from 320 to 1080 nm, Baltic Astronomy, 5, 603-838, 1996.

Arnoldy, R. J., and P. B. Lewis, Jr, Correlation of ground-based and topside photometric observations with auroral electron spectra measurements at rocket altitudes, Tech. rep., Space Science Center University of New Hampshire, Space Science Center University of New Hampshire Durham, NH 03824, 1976.

Ångstrom, A. J., Researches sur le Spectre Solaire, W. Schultz, Uppsala, 1868.

Ångstrom, A.J., Spectrum des Nordlichts, Pogg. Ann., 137, 161-163, 1869.

Backer, D.J., and G.J.Romick, The Rayleigh interpretation of the unit in terms of column emission rate or apparent radiance expressed in SI units, Applied Optics, 15, 1966-1968, 1976.

Bak T. and R.Wisniewski and M. Blanke, Autonomous attitude determination and control system for the rsted satellite, In IEEE Aerospace Apllications Conference, February 1996.

Brändström U, IRF Scientific Report 179, 2003.

Gustavsson B, Three Dimensional Imaging of Aurora and Airglow, PhD thesis, 2000.

Henricson H. Results from the intercalibration of low light level sources at Andoya 2007, Proceeding of the 33rd Annual European Meeting on Atmospheric Studies by Optical Methods, 131, 2008.

Herzberg, L. Solar optical radiation and its role in upper atmospheric processes. In C. O. Hines, I. Paglis, T. R. Hartz, and J. A. Fejer, editors, physics of the Earth's Upper Atmosphere, pages 31-45. Prentice Hall. 1965

Howell. S. B. Astronomical CCD observing and reduction techqiques Astronomical Society of Pacific (ASP Conference Series. 23, 345.

Hunten, D.M., F. E. Roach, and J. W. Chamberlain, A photometric unit for the aurora and airglow, J.Atmos. Terr. Phys., 8, 345-346, 1956.

Kitchin, Christopher Robert. Stars, Nebulae and the Interstellar Medium: Observational Physics and Astrophysics. ORC press. 1987.

Klaus A and Bauer J, Camera Calibration from a Single Night Sky Image, Proceeding of the 2004 IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2004.

Lauche, H., and O. Widell, Intercalibration of low light level sources, in Proc.of 27th Annual European Meeting on Atmospheric Studies by Optical Methods,Stockholm, Sweden, Meteorological institution, Stockholm university, Sweden,2000b.

Massey, P., A User's Guide to CCD Reductions with IRAF, National Optical Astronomy Observatory, 1992.

Meier, R. R.(1991). Ultraviolet spectroscopy and remote sensing of the upper atmosphere. Apace Sci. Rev., 58, 1- 185.

Pulkovo spectrophotometric catalogue, Baltic Astronomy ,5 ,4, 1996.

Romick, G.J., and A. E. Belon, The spatial variation of auroral luminosity-II, Planet Space Sci., 15, 1695-1716, 1967.

Rufino G, Accardo D. Enhancement of the centroiding algorithm for star tracker measure refinement, Acta Astronautica ,135-147, 2003.

Schmid H.H., Stellar Calibration of the Orbigon lens, Photogrammetric Engineering, 40(1), 101-111, 1974.

Solomon, S.C.,P.B. Hays, and V.K.Abreu, Tomographic inversion of satellite photometry, Applied Optics, 23, 3409-3414, 1984.

Steen, A., An Auroral Large Imaging System in Northern Scandinavian, in Proc. 9th Symposium on European rocket and ballon programmes and related research, Lahnstein, FRG, pp.299-303, European Space Agency, 1989, ESA SP-291.

Steen, A., and U. Brändström, A multi-station ground-based imaging system at high latitudes, STEP Int.Newsl., 3, 11-14, 1993.

West J., Reducing astronomical data using ImageJ (Univ. Manitoba), http://www.umanitoba.ca/faculties/science/astronomy/jwest/reduction_recipe.pdf.

Widell, O., and H.Henricson, Intercalibration of low light level sources, in Proc. Of 28th Annual European Meeting on Atmospheric Studies by Optical Methods, Oulu, Finland, Sodankyla Geophysical Observatory, 2001.

Wisniewski R. and M. Blank, Three-axis Satellite Attitude Control Based on Magnetic Torquing, 13th IFA C World Congress, June 1996.